QUANT SERIES · PART 3 OF 5

Stochastic Calculus, Step by Step

Part 3: Brownian motion and the Itô integral



Rates with Bip

STOCHASTIC CALCULUS, STEP BY STEP · PART 3 OF 5

Stochastic Calculus, Step by Step

Part 3: Brownian motion and the Itô integral, with Python

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Stochastic Calculus, Step by Step. Part 3: Brownian motion and the Itô integral, with Python.

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One set of numbers runs through the whole course: a share priced at $S_0 = 100$, a strike $K = 105$, a risk-free rate $r = 3\%$ (continuously compounded), a volatility $\sigma = 20\%$ and a horizon $T = 1$ year; in Part 3 the share's expected return is taken equal to r, $\mu = 3\%$. They are made up for teaching, and every random path is simulated with a fixed seed. There is no market data.

The exercises are at the end of each lesson. Their full solutions, and the answers to the self-test questions, are in the companion volume, Stochastic Calculus, Step by Step, Part 3: Solutions.

Every number in the text, tables and figures is computed by the companion Python notebooks (numpy only), one per lesson, which reproduce each result.

This book is for education only. It is not financial, investment, legal or tax advice.

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How to use this book

This is Part 3 of *Stochastic Calculus, Step by Step*, a course in five parts that goes from school calculus to the Black-Scholes formula, Feynman-Kac and the numerical methods banks use to price options. Part 1 laid the mathematical foundations; Part 2 built option pricing on a coin-toss tree. Part 3 lets the tree's steps shrink to nothing. The coin tosses become **Brownian motion**, sums of trading gains become the **Itô integral**, and the chain rule picks up an extra term: **Itô's lemma**. This is the calculus the rest of the course is written in.

LESSON	TOPIC	USED LATER FOR
13	Brownian motion: the limit of a random walk, its covariance, three martingales	every continuous-time model
14	No slope, infinite length, quadratic variation T; $dW\,dW = dt$	Itô's lemma, hedging error
15	The Itô integral: left endpoints, martingale, isometry; $\int W\,dW$	gains from trading, the BS PDE
16	Itô's lemma, derived from Taylor and $dW^2 = dt$	everything that follows
17	Geometric Brownian motion: solving the share's equation; Euler's method	the Black-Scholes model, simulation
18	Spotting martingales; the exponential martingale; the product rule	risk-neutral pricing, Girsanov

Who this book is for

As in Parts 1 and 2, this book is written for readers learning on their own, without a teacher to fill the gaps. Every new idea is **defined** before it is used, **explained** in words, tried on a small example, and then **derived** one line at a time, with the reason for each line. When a step uses an earlier lesson, the lesson is named.

You need Part 1's calculus and probability, above all Taylor's theorem (Lesson 1), the normal distribution and its moments (Lesson 4), the lognormal (Lesson 5) and the convergence argument "mean to a constant, variance to zero" (Lesson 6). From Part 2 you need conditional expectation and martingales (Lessons 8 and 11).

One set of numbers

Part 3 returns to the course's numbers: a share at $S_0 = 100$, a volatility $\sigma = 20\%$, a risk-free rate $r = 3\%$ (continuously compounded), a strike $K = 105$ and a horizon of $T = 1$ year. Lesson 17 needs the share's expected return μ as well; we take $\mu = r = 3\%$ there, so that every result can be checked against Lesson 5. Every random path is simulated with a fixed seed.

How each lesson is built

Every lesson follows the same steps as before: **why this lesson**, **what you need** from earlier, **the picture**, the derivations as numbered steps, **numbers and code**, **what to remember**, **common mistakes**, eight **exercises** (six by hand, two in code) and a three-question **self-test**.

How to read it

Part 3 is where most self-taught readers get stuck, usually at one of three places: why Brownian motion has no derivative, why the integral must use left endpoints, and where the $\frac{1}{2}\sigma^2$ in Itô's lemma comes from. Each of these is explained twice here, once with algebra and once with a simulation you can run. If an argument doesn't click, run the notebook, change the number of steps, and watch the quantity in question settle (or refuse to).

Notebooks and solutions

Every number, table and figure is computed by a companion Python notebook, one per lesson, using numpy only. The notebooks are free at github.com/RatesWithBip/curriculum.

The solutions are **not** in this book: they are in the companion volume, *Part 3: Solutions*, with every derivation written out and every code solution shown with what it prints. Try each exercise before you look.

Reading alongside

Steven Shreve's *Stochastic Calculus for Finance II*, Chapters 3 and 4, covers the same ground with full rigour; Thomas Mikosch's *Elementary Stochastic Calculus with Finance in View*, Chapters 1 to 3, is gentler and close in spirit to this book. This book is written fresh from its own notebooks and uses none of their examples or exercises.

Free, and staying free

The videos, the notebooks and both books are free. If they help you, you can support the channel with a coffee at ko-fi.com/rateswithbip.

This book is for education only and is not financial advice.

LESSON 13 · VIDEO 13

Brownian Motion

How a coin-toss walk with ever smaller, ever faster steps becomes Brownian motion, and why the steps must shrink like one over the square root of their number; the four properties that define it, checked on simulated paths; its covariance, min(s, t); self-similarity, seen by zooming in; and the three martingales built from it, W, $W^2 - t$ and the exponential martingale, which makes the discounted share price a fair game.

Why this lesson

Part 2 priced options on a tree whose share moved every four months. Real prices move all the time. To model that, we let the tree's steps get shorter and more numerous until the path moves continuously. The limit is **Brownian motion**, written W_t: the random ingredient of every continuous-time model in finance.

- The Black-Scholes model drives the share with it: $S_t = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}$ (Lesson 17).
- The Itô integral (Lesson 15) and Itô's lemma (Lesson 16) are calculus for functions of it.
- Girsanov's theorem (Lesson 25) changes its drift by reweighting its paths.

This lesson builds it from a random walk, states its definition, works out its covariance, looks at its strange self-similar shape, and finds three martingales hidden in it.

What you need from earlier lessons

- **Normal distributions** (Lesson 4). If $X \sim N(0, v)$, then $E[X] = 0$, $E[X^2] = v$, and $E[e^{uX}] = e^{u^2 v/2}$. Sums of independent normals are normal, with variances adding.
- **The central limit theorem** (Lesson 5). A sum of many small independent pieces is close to normal.
- **Covariance and independence** (Lesson 3). $\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y]$; independent variables have covariance 0, and $E[XY] = E[X]E[Y]$.
- **Conditional expectation and martingales** (Lessons 8 and 11). Taking out what is known; independence; $E[M_t \mid \mathcal{F}_s] = M_s$.

The picture

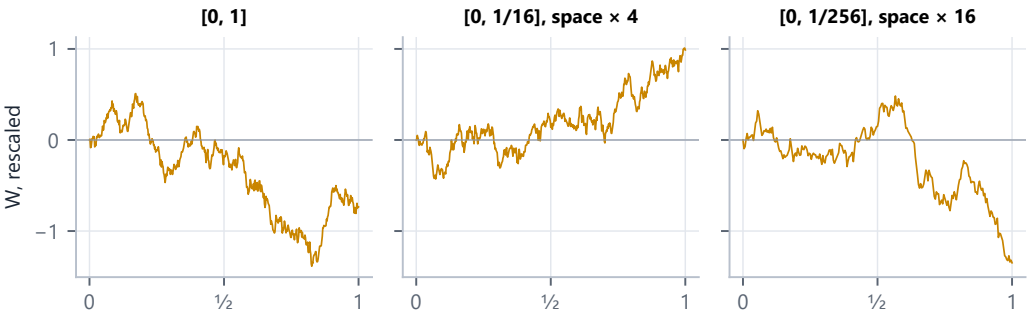


Figure 13.1. One simulated Brownian path on [0, 1] (left), and its first sixteenth (middle) and first 256th (right), each stretched to fill the panel, with the vertical scale multiplied by 4 and by 16. The zoomed pieces look just like the whole: a Brownian path is equally rough at every scale.

Zoom into a smooth curve and it straightens out into a line; that is what a derivative is. Zoom into a Brownian path and you see another Brownian path. The rule for zooming is the key to everything in Part 3: stretching time by 16 stretches space by only $\sqrt{16} = 4$.

From a random walk to Brownian motion

The scaled walk

Toss a fair coin n times per unit of time. After each toss the walker moves up or down by the same step size δ. After time t there have been nt tosses (take nt to be a whole number), so the walker's position is

$$W_t^{(n)} = \delta(X_1 + X_2 + \cdots + X_{nt}), \qquad X_i = \pm 1 \text{ with probability } \tfrac{1}{2} \text{ each.}$$

Step 1. The mean. Each X_i has mean $\frac{1}{2}(1) + \frac{1}{2}(-1) = 0$, so $E[W_t^{(n)}] = 0$.

Step 2. The variance. Each X_i has $X_i^2 = 1$, so variance 1. The tosses are independent, so variances add (Lesson 3): $\mathrm{Var}(W_t^{(n)}) = \delta^2 \times nt$.

Step 3. Choosing the step size. We want the walk to settle on something as n grows. Look at the variance $\delta^2 nt$:

- with $\delta = 1$ it is nt, which blows up: the walk wanders off to infinity;
- with $\delta = 1/n$ it is t/n, which goes to 0: the walk collapses to a flat line;
- with $\delta = 1/\sqrt{n}$ it is exactly t, **whatever n is**.

Only $\delta = 1/\sqrt{n}$ gives a non-trivial limit. This square root is the signature of everything random in this course: over a time Δt, a random path moves about $\sqrt{\Delta t}$, not Δt.

Step 4. The limit. $W_1^{(n)}$ is a sum of n independent pieces $\pm 1/\sqrt{n}$, with mean 0 and variance 1. By the central limit theorem it becomes $N(0,1)$ as $n \to \infty$. The notebook simulates the walk at time 1 for several n:

The scaled walk at time 1, 400,000 simulations each. N(1) = 0.8413.

STEPS PER UNIT TIME	MEAN	VARIANCE	P(walk ≤ 1)
10	-0.0046	1.0019	0.8293
100	0.0008	1.0005	0.8638
1,000	0.0031	0.9967	0.8361
10,000	-0.0012	1.0004	0.8437

The variance is 1 for every n, as Step 2 promised. The probability of ending at or below 1 approaches $N(1) = 0.8413$, though not smoothly: for small n the walk can only land on a few values (spaced $2/\sqrt{n}$ apart), and whether one of them sits exactly at 1 matters. The picture shows the whole distribution function settling on the normal one:

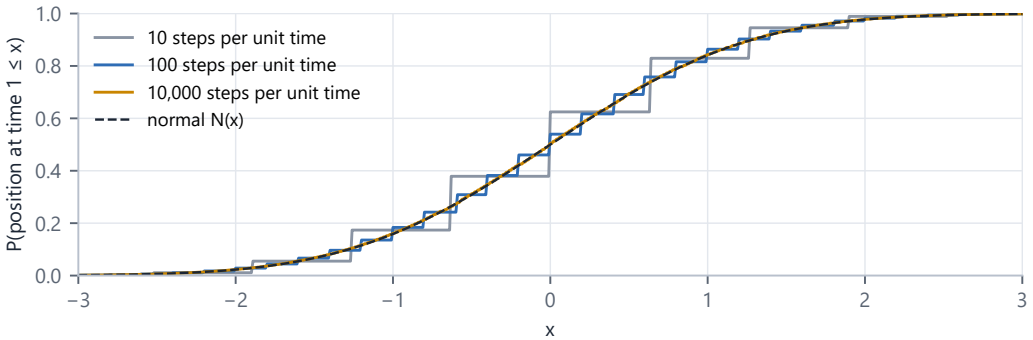


Figure 13.2. P(walk at time 1 ≤ x) for 10, 100 and 10,000 steps per unit of time, against the normal distribution function N(x). With few steps it is a staircase; with many, the staircase melts into the smooth normal curve.

Step 5. What else carries over. Over a time interval from s to t, the walker's move is made of the tosses in that interval only, so moves over separate intervals are independent; and each move is a sum of many small pieces, so it becomes normal with variance $t - s$. These are exactly the properties in the definition below. (A theorem of Donsker makes "the walk converges to Brownian motion" precise, path by path; we will not need its proof.)

The definition

Definition. A **Brownian motion** (or **Wiener process**) is a family of random variables W_t, $t \geq 0$, such that

1. **It starts at 0:** $W_0 = 0$.
2. **Independent increments:** for times $0 \leq t_0 < t_1 < \cdots < t_k$, the moves $W_{t_1} - W_{t_0},\ W_{t_2} - W_{t_1}, \ldots, W_{t_k} - W_{t_{k-1}}$ are independent.
3. **Normal increments:** for $s < t,\ W_t - W_s \sim N(0, t-s)$.
4. **Continuous paths:** for each outcome, $t \mapsto W_t$ is a continuous function.

Read each one:

1. A convention: we measure moves from the start.
2. The future moves have nothing to do with the past ones. This is the coin tosses' independence, carried over. It makes W a Markov process (Lesson 11) and gives every martingale below.
3. The size of a move depends only on **how long** you wait, not on when you start, and its variance equals the waiting time. In particular $W_t \sim N(0, t)$, with standard deviation $\sqrt{t}$.
4. No jumps. (Models with jumps come in Part 5.)

Simulating it. Properties 2 and 3 tell us how: cut $[0, 1]$ into n steps of length Δt, draw independent $N(0, \Delta t)$ moves, and add them up. The notebook does this for 20,000 paths of 1000 steps and checks the definition:

TIME t	AVERAGE OF W_t	VARIANCE OF W_t
0.25	0.0000	0.2492
0.5	-0.0028	0.4936
1	0.0051	1.0126

The variance grows in step with time. The moves over $[0.25, 0.5]$ and $[0.5, 1]$ have correlation 0.0005, zero up to simulation noise, and the second has variance 0.5096, close to $1 - 0.5 = 0.5$.

Brownian paths: add up independent normal steps.

```
n_paths, n_steps = 20_000, 1000
dt = 1.0 / n_steps
dW = rng.normal(0.0, math.sqrt(dt), (n_paths, n_steps))
```

Covariance: min(s, t)

How related are the values of the path at two different times?

Claim. For any $s, t \geq 0$, $\operatorname{Cov}(W_s, W_t) = \min(s, t)$.

Proof. Say $s \leq t$.

Step 1. Split the later value. $W_t = W_s + (W_t - W_s)$: where the path was at time s, plus the move since.

Step 2. Multiply and average. Both have mean 0, so the covariance is $E[W_s W_t]$:

$$E[W_s W_t] = E[W_s^2] + E\big[W_s (W_t - W_s)\big].$$

Step 3. Use the definition. $E[W_s^2] = \operatorname{Var}(W_s) = s$. The move $W_t - W_s$ is independent of $W_s = W_s - W_0$ (property 2), so $E[W_s (W_t - W_s)] = E[W_s]\, E[W_t - W_s] = 0 \times 0 = 0$.

Step 4. So $\operatorname{Cov}(W_s, W_t) = s = \min(s, t)$.

Meaning. The two values share the whole of the path up to the earlier time, and nothing after it. The correlation is $\min(s,t)/\sqrt{st} = \sqrt{s/t}$ for $s \leq t$: high when the times are close, low when the later time is much later.

Covariance and correlation of the path at two times.

EARLIER TIME	LATER TIME	Simulated $E[W_s W_t]$	min(s, t)	SIMULATED CORRELATION	√(s/t)
0.25	0.75	0.2491	0.25	0.5766	0.5774
0.50	1.00	0.4983	0.50	0.7048	0.7071
0.30	0.60	0.2973	0.30	0.7065	0.7071
0.80	0.90	0.8015	0.80	0.9429	0.9428

Example. $\operatorname{Var}(W_1 + W_2) = \operatorname{Var}(W_1) + \operatorname{Var}(W_2) + 2\operatorname{Cov}(W_1, W_2) = 1 + 2 + 2 \times 1 = 5$. (Not $1 + 2 = 3$: the two values are positively related.)

Zooming in: self-similarity

Claim. For any $c > 0$, the process $B_t = \dfrac{1}{\sqrt{c}} W_{ct}$ is again a Brownian motion.

Why. Check the four properties. $B_0 = W_0/\sqrt{c} = 0$. Moves of B over separate intervals are moves of W over separate (stretched) intervals, so they are independent. For $s < t$, $B_t - B_s = (W_{ct} - W_{cs})/\sqrt{c}$, which is normal with mean 0 and variance $(ct - cs)/c =$

$t - s$. And B is continuous because W is. (Exercise 13.4 writes this out.)

Meaning. Take the path on $[0, 1/c]$, stretch time by c to fill $[0, 1]$, and multiply space by $\sqrt{c}$: you get something with exactly the same statistics as the original path. That is the picture at the top of the lesson, with $c = 16$ and $c = 256$. A smooth curve, zoomed by 256 in time, would need its height multiplied by 256, not 16, to stay visible; a Brownian path needs only $\sqrt{256} = 16$. In the notebook's path, the range of the three panels is 1.98, 1.50 and 1.83: about the same at every scale.

This is the first hint of Lesson 14's result: a Brownian path is too rough to have a slope anywhere.

Three martingales

Let $\mathcal{F}_s$ be the information in the path up to time s: the continuous-time filtration of Lesson 7. A process M_t is a **martingale** if $E[M_t \mid \mathcal{F}_s] = M_s$ for all $s \le t$ (Lesson 11). Every proof below uses the same two moves:

- **split**: $W_t = W_s + D$, with $D = W_t - W_s \sim N(0, t - s)$;
- **independence**: D is independent of $\mathcal{F}_s$, so $E[g(D) \mid \mathcal{F}_s] = E[g(D)]$, while anything built from W_s is known at time s and comes out (Lesson 8).

1. W itself

$$E[W_t \mid \mathcal{F}_s] = E[W_s + D \mid \mathcal{F}_s] = W_s + E[D] = W_s.$$

A fair game: the best forecast of where the path will be is where it is now.

2. $W^2 - t$

Step 1. Square the split. $W_t^2 = W_s^2 + 2W_sD + D^2$.

Step 2. Condition on $\mathcal{F}_s$. W_s^2 is known; $E[2W_sD \mid \mathcal{F}_s] = 2W_s\,E[D] = 0$; $E[D^2 \mid \mathcal{F}_s] = E[D^2] = t - s$. So

$$E[W_t^2 \mid \mathcal{F}_s] = W_s^2 + (t - s).$$

Step 3. Subtract t. $E[W_t^2 - t \mid \mathcal{F}_s] = W_s^2 - s$.

So W_t^2 on its own is **not** a martingale: it drifts upwards at rate 1, because the path spreads out. Subtracting t removes exactly that drift. (Lesson 18 redoes this in one line with Itô's lemma.)

3. The exponential martingale

For a constant σ, let $Z_t = e^{\sigma W_t - \frac{1}{2}\sigma^2 t}$.

Step 1. Split. $Z_t = e^{\sigma W_s - \frac{1}{2}\sigma^2 t}\, e^{\sigma D}$.

Step 2. Condition. The first factor is known at time s. For the second, $D \sim N(0, t-s)$, so by Lesson 4, $E[e^{\sigma D}] = e^{\frac{1}{2}\sigma^2 (t-s)}$. So

$$E[Z_t \mid \mathcal{F}_s] = e^{\sigma W_s - \frac{1}{2}\sigma^2 t}\, e^{\frac{1}{2}\sigma^2 (t-s)} = e^{\sigma W_s - \frac{1}{2}\sigma^2 s} = Z_s.$$

The $-\frac{1}{2}\sigma^2 t$ is exactly what is needed: $e^{\sigma W_t}$ alone would drift upwards (by Jensen's inequality, Lesson 5), and the factor $e^{-\frac{1}{2}\sigma^2 t}$ takes the drift out. $E[Z_t] = Z_0 = 1$ for every t.

The share is a fair game, after discounting

Let the share be $S_t = S_0 e^{(r - \frac{1}{2}\sigma^2)t + \sigma W_t}$, the lognormal model of Lesson 5 with drift r. Then

$$e^{-rt} S_t = S_0 e^{\sigma W_t - \frac{1}{2}\sigma^2 t} = S_0 Z_t,$$

a martingale. The discounted share price is a fair game, just as on the tree in Lesson 11. In Part 4 this is what makes the probability under which W is a Brownian motion a **risk-neutral** one.

Checking on simulated paths

Averages at time 1 should equal the starting values: $E[W_1] = 0.0051$ (0), $E[W_1^2 - 1] = 0.0126$ (0), $E[Z_1] = 1.0013$ (1, with $\sigma = 20\%$), and $E[e^{-r} S_1] = 100.13$ (100).

A martingale says more: the forecast is right **given any information at time s**, not just on average. The notebook sorts the 20,000 paths into ten groups by their value at time 0.5 and compares, in each group, the average at time 1 with the average at time 0.5:

Paths grouped by where they are at time 0.5. In each group the time-1 average matches the time-0.5 value.

GROUP	Average $W_{0.5}$	$W^2 - t$ at 0.5	$W^2 - t$ at 1	Z AT 0.5	Z AT 1
1	-1.232	1.100	1.125	0.7751	0.7735
2	-0.741	0.056	0.082	0.8539	0.8543
3	-0.480	-0.265	-0.258	0.8995	0.8996
4	-0.276	-0.421	-0.447	0.9369	0.9415
5	-0.093	-0.489	-0.455	0.9719	0.9757
6	0.088	-0.489	-0.504	1.0077	1.0037
7	0.271	-0.424	-0.412	1.0452	1.0509
8	0.472	-0.273	-0.242	1.0881	1.0929
9	0.733	0.045	0.051	1.1464	1.1452
10	1.230	1.097	1.185	1.2683	1.2757

The largest gaps are 0.088 for $W^2 - t$ and 0.0074 for Z: simulation noise, with about 2,000 paths per group.

What to remember

- A walk with n steps of size $\pm 1/\sqrt{n}$ per unit time has variance t at time t for every n; as $n \to \infty$ it becomes Brownian motion.
- **Brownian motion:** $W_0 = 0$; independent increments; $W_t - W_s \sim N(0, t-s)$; continuous paths. So $W_t \sim N(0, t)$.
- Over a time Δt a random path moves about $\sqrt{\Delta t}$.
- $\mathrm{Cov}(W_s, W_t) = \min(s, t)$; correlation $\sqrt{s/t}$ for $s \le t$.
- **Self-similarity:** $W_{ct}/\sqrt{c}$ is again a Brownian motion.
- **Martingales:** $W_t,\ W_t^2 - t,\ e^{\sigma W_t - \frac{1}{2}\sigma^2 t}$. Proof pattern: split $W_t = W_s + (W_t - W_s)$, use independence.
- With $S_t = S_0 e^{(r - \frac{1}{2}\sigma^2)t + \sigma W_t}$, the discounted price $e^{-rt} S_t$ is a martingale.

Common mistakes

- **Steps of size Δt.** A random walk scaled by $1/n$ collapses to 0. Random moves over a time Δt are of size $\sqrt{\Delta t}$.
- **"W_t has standard deviation t."** Its *variance* is t; its standard deviation is $\sqrt{t}$.
- **"W_s and W_t are independent."** The *increments* over separate intervals are independent; the values share the path up to the earlier time, so their covariance is

$\min(s,t)$.

- **"W_t^2 is a martingale."** It drifts up at rate 1; $W_t^2 - t$ is the martingale.
- **"$e^{\sigma W_t}$ is a martingale."** It drifts up too ($E[e^{\sigma W_t}] = e^{\frac{1}{2}\sigma^2 t}$). The $-\frac{1}{2}\sigma^2 t$ in the exponent is essential.
- **"The share price is a martingale."** Only after discounting, and only when its drift is r.

Exercises

Solutions are in *Part 3: Solutions*.

13.1 A walker takes n steps per unit of time, each $+1/\sqrt{n}$ or $-1/\sqrt{n}$ with probability $\frac{1}{2}$. Show that at time t (with nt a whole number) its position has mean 0 and variance t. Why do the steps have to be of size $1/\sqrt{n}$, and not $1/n$?

13.2 Prove that $\mathrm{Cov}(W_s, W_t) = \min(s,t)$. Use it to find $\mathrm{Var}(W_1 + W_2)$ and the correlation of W_1 and W_4.

13.3 Find $P(W_1 > 0 \text{ and } W_2 > W_1)$. You are told that $P(W_1 > 0 \text{ and } W_2 > 0) = \frac{3}{8}$. Why are the two different?

13.4 Let $c > 0$ and $B_t = W_{ct}/\sqrt{c}$. Check that B has the four properties of a Brownian motion.

13.5 Prove that $W_t^2 - t$ is a martingale. (Write $W_t = W_s + (W_t - W_s)$ and expand the square.)

13.6 Prove that $Z_t = e^{\sigma W_t - \frac{1}{2}\sigma^2 t}$ is a martingale, using $E[e^{uX}] = e^{u^2 v/2}$ for $X \sim N(0, v)$ (Lesson 4). Deduce that $e^{-rt}S_t$ is a martingale when $S_t = S_0 e^{(r - \frac{1}{2}\sigma^2)t + \sigma W_t}$.

13.7 *(Code)* Simulate scaled random walks with $n = 10$, 100 and 10,000 steps per unit of time, 200,000 times each, and compare $P(W_1^{(n)} \le 1)$ with $N(1)$. Why is the answer for small n so far off?

13.8 *(Code)* Simulate 50,000 Brownian paths on $[0, 2]$ and check $\mathrm{Cov}(W_s, W_t) = \min(s,t)$ on a grid of pairs, and $E[Z_t] = 1$ for $t = 0.5, 1, 2$ with $\sigma = 0.2$.

Self-test

1. What is the variance of $W_{0.25}$? Its standard deviation?
2. Why must the steps of the random walk shrink like $1/\sqrt{n}$?
3. Which of W_t, W_t^2, e^{W_t} are martingales? Fix the ones that aren't.

Companion notebook: `s5e13_brownian_motion.ipynb`. Every number in this chapter is computed by it.

LESSON 14 · VIDEO 14

The Strange Properties of Brownian Paths

Why a Brownian path has no slope at any time; why it is infinitely long (its first variation diverges); why the sum of its squared moves settles on exactly the time elapsed (quadratic variation T), path by path; the multiplication table dW·dW = dt, dW·dt = 0, dt·dt = 0 that Itô calculus runs on; and the difference between E[W²] = T, an average over paths, and [W] = T, a fact about every path.

Why this lesson

Ordinary calculus is built for smooth curves: curves with a slope at every point, whose length you can measure. Brownian motion is not such a curve. This lesson shows exactly how it fails, and finds the one property that replaces the slope:

- the slope $(W_{t+h} - W_t)/h$ does not settle as $h \to 0$: there is **no derivative**;
- the total distance travelled, $\sum |\Delta W|$, is **infinite**;
- but the sum of the **squared** moves, $\sum (\Delta W)^2$, settles on the elapsed time T, on every path.

That last fact, written $dW\,dW = dt$, is the engine of Itô's lemma (Lesson 16), of the Black-Scholes equation (Lesson 19) and of every hedging-error calculation in Part 4. Lesson 1 met it as a surprise and Lesson 6 proved it on average; here it becomes a property of the path itself.

What you need from earlier lessons

- **Brownian motion** (Lesson 13). $W_0 = 0$; independent increments; $W_t - W_s \sim N(0, t-s)$; continuous paths.
- **Moments of the normal** (Lesson 4). For $Z \sim N(0,1)$: $E[Z^2] = 1$, $E[Z^4] = 3$, $E|Z| = \sqrt{2/\pi}$, and $P(|Z| > a) = 2(1 - N(a))$.
- **Mean to a constant, variance to zero** (Lesson 6). If $E[X_n] \to c$ and $\mathrm{Var}(X_n) \to 0$, then $X_n \to c$.

- **Derivatives** (Lesson 1). $f'(t) = \lim_{h\to 0} \big(f(t+h) - f(t)\big)/h$.

Notation. Cut $[0, T]$ into n equal steps of length $\Delta t = T/n$, at times $t_i = i\Delta t$. The moves are $\Delta W_i = W_{t_{i+1}} - W_{t_i}$: independent, each $N(0, \Delta t)$, so each is $\sqrt{\Delta t}$ times a standard normal.

The picture

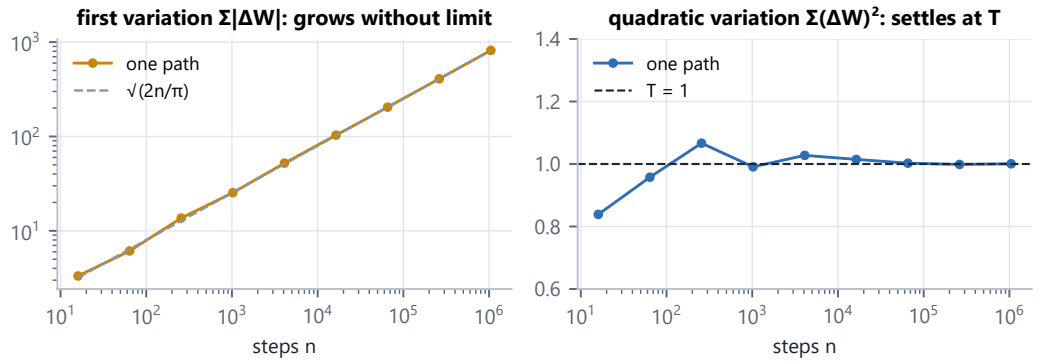


Figure 14.1. One Brownian path on [0, 1], cut into n steps for n from 16 to about a million. Left: the total distance travelled, Σ|ΔW|, grows like √(2n/π) without limit (both axes logarithmic). Right: the sum of squared moves, Σ(ΔW)², settles on T = 1.

The same path, measured two ways. Its **length** keeps growing as you look more closely, because every zoom reveals more wiggles (Lesson 13's self-similarity). Its **sum of squares** stops changing. By the end of the lesson you will be able to compute both curves.

No slope anywhere

Step 1. The slope over a step h. $Q_h = \dfrac{W_{t+h} - W_t}{h}$. The numerator is $N(0, h)$, so dividing by h:

$$Q_h \sim N\Big(0, \frac{h}{h^2}\Big) = N\Big(0, \frac{1}{h}\Big), \qquad \mathrm{sd}(Q_h) = \frac{1}{\sqrt{h}}.$$

Step 2. Let h shrink. For a derivative to exist, Q_h would have to settle on a number. Instead its spread **grows**: the standard deviation $1/\sqrt{h}$ goes to infinity. For any fixed bound M,

$$P(|Q_h| > M) = P\big(|Z| > M\sqrt{h}\big) = 2\big(1 - N(M\sqrt{h})\big) \to 2(1 - N(0)) = 1.$$

As $h \to 0$ the slope is almost certain to exceed any bound you name.

Numbers, with $M = 10$:

The slope of a Brownian path over shrinking steps: it never settles.

STEP h	SIMULATED SD OF SLOPE	$1/\sqrt{h}$	P(|slope| > 10)	FORMULA
0.1000	3.16	3.16	0.0014	0.0016
0.0100	10.01	10.00	0.3170	0.3173
0.0010	31.61	31.62	0.7510	0.7518
0.0001	99.95	100.00	0.9206	0.9203

This shows the slope fails to exist at any **given** time. A stronger theorem (Paley, Wiener and Zygmund, 1933) says that, with probability 1, a Brownian path has no derivative at **any** time at all. We will not prove it; the table already shows why it is true: the square root again. A move over time h is of size $\sqrt{h}$, and $\sqrt{h}/h = 1/\sqrt{h}$ blows up.

First variation: infinitely long

Definition. The **first variation** of a path over $[0, T]$ is the limit of $\sum_i |\Delta W_i|$ as the steps shrink: the total distance travelled, up and down. For a smooth path with slope f' it is $\int_0^T |f'(t)|\,dt$, a finite number.

Step 1. One move. $\Delta W_i = \sqrt{\Delta t}\, Z_i$, so $E|\Delta W_i| = \sqrt{\Delta t}\, E|Z| = \sqrt{\Delta t}\sqrt{2/\pi}$ (Lesson 4).

Step 2. Add up n of them.

$$E\sum_{i=1}^{n} |\Delta W_i| = n\sqrt{\frac{T}{n}}\sqrt{\frac{2}{\pi}} = \sqrt{\frac{2nT}{\pi}}.$$

Step 3. Let n grow. This grows like $\sqrt{n}$, without limit. (Its standard deviation is only about $\sqrt{T(1-2/\pi)}$, so the sum is reliably close to its mean: the path really is that long.)

The left panel of the picture shows one path: from about 3.3 with 16 steps to 817 with a million, tracking $\sqrt{2n/\pi}$.

Why this matters. Ordinary (Riemann-Stieltjes) integrals $\int H\,dW$ are defined for paths of finite length. Brownian paths are infinitely long, so Lesson 15 needs a new kind of integral.

Quadratic variation: exactly T

Definition. The **quadratic variation** of a path over $[0, T]$, written $[W]_T$, is the limit of $\sum_i (\Delta W_i)^2$ as the steps shrink.

For a smooth path it is 0. If f has a slope bounded by M, each move is at most $M\Delta t$, so

$$\sum_i (\Delta f_i)^2 \le n(M\Delta t)^2 = M^2 T\,\Delta t \to 0.$$

Squares of small numbers are much smaller, and there aren't enough of them to add up to anything.

For Brownian motion it is T.

Step 1. The mean. $E[(\Delta W_i)^2] = \Delta t$, so $E\sum (\Delta W_i)^2 = n\Delta t = T$.

Step 2. The variance. $(\Delta W_i)^2 = \Delta t\, Z_i^2$ has variance $\Delta t^2 \operatorname{Var}(Z^2) = \Delta t^2 (E[Z^4] - E[Z^2]^2) = \Delta t^2 (3 - 1) = 2\Delta t^2$. The moves are independent, so variances add:

$$\operatorname{Var} \sum_i (\Delta W_i)^2 = n \cdot 2\Delta t^2 = \frac{2T^2}{n} \to 0.$$

Step 3. Converge. Mean T, variance to zero: by Lesson 6, $\sum (\Delta W_i)^2 \to T$.

$$\boxed{[W]_T = T}$$

Here the squares are of size Δt, not Δt^2, and there are exactly enough of them ($n = T/\Delta t$) to add up to T.

On the notebook's path:

First and quadratic variation of one path, cut ever more finely (T = 1).

STEPS N	Σ|ΔW|	√(2n/π)	Σ(ΔW)²	Its sd, √(2/n)
16	3.33	3.19	0.8388	0.3536
64	6.12	6.38	0.9577	0.1768
256	13.65	12.77	1.0666	0.0884
1,024	25.37	25.53	0.9909	0.0442
4,096	52.16	51.06	1.0276	0.0221
16,384	103.36	102.13	1.0147	0.0110
65,536	204.44	204.26	1.0026	0.0055
262,144	408.24	408.52	0.9987	0.0028
1,048,576	817.43	817.03	1.0007	0.0014

For the share: the volatility is visible on one path

Apply the same idea to the log price, $\ln S_t = \ln S_0 + (r - \frac{1}{2}\sigma^2)t + \sigma W_t$. Its moves are $\sigma \Delta W_i$ plus a drift of order Δt; the drift's squares vanish, so $\sum (\Delta \ln S)^2 \to \sigma^2 T$. On one simulated year:

STEPS	16	256	4,096	65,536	1,048,576
$\sum (\Delta \ln S)^2$	0.0335	0.0427	0.0411	0.0401	0.0400

against $\sigma^2 T = 0.04$. This is **realised variance**, how traders measure volatility from price data. In theory one path, watched closely enough, reveals σ exactly. The expected return, by contrast, cannot be pinned down from one path however closely you watch, a fact that matters in Lesson 19.

For comparison, a smooth path (cash growing at 3% from 100) has $\sum (\Delta f)^2 = 0.5797$, 0.009058 and 1.4e-04 with 16, 1,024 and 65,536 steps: heading to 0.

The multiplication table

Itô calculus keeps track of products of small moves over $[0, T]$, added up over the steps. Three sums cover every case:

$\sum \Delta t^2$. It is $n \Delta t^2 = T \Delta t \to 0$.

$\sum \Delta W_i \, \Delta t$. Its mean is 0 and its variance is $\Delta t^2 \sum \mathrm{Var}(\Delta W_i) = \Delta t^2 \cdot n \Delta t = T \Delta t^2 \to 0$. So it goes to 0.

$\sum (\Delta W_i)^2$. It goes to T.

Written as a table of "products of differentials", with the understanding that each entry is what the sum of such products tends to:

$\times$	dt	dW
dt	0	0
dW	0	dt

The rule of thumb. dt is of order Δt and dW is of order $\sqrt{\Delta t}$. A product of total order higher than Δt (like dt^2, or $dW\,dt \sim \Delta t^{3/2}$) adds up to 0; a product of order exactly Δt (dW^2) adds up to something finite; and dW alone (order $\sqrt{\Delta t}$) adds up to a random, finite quantity.

The multiplication table on one path.

STEPS N	Σ Δt²	Σ ΔW Δt	Σ ΔW²
64	0.015625	-0.003500	0.9577
1,024	0.000977	-0.000219	0.9909
16,384	0.000061	-0.000014	1.0147
262,144	0.000004	-0.000001	0.9987

Two Brownian motions. If B is another Brownian motion with correlation ρ to W (built as $B = \rho W + \sqrt{1-\rho^2}\, W'$ with W' independent), then $dW\, dB = \rho\, dt$; for independent ones, $dW\, dB = 0$. On the notebook's path with $\rho = 0.6$: $\sum \Delta W\, \Delta B = 0.6011$; with an independent one, 0.0005. (Exercise 14.6.)

Average squares versus path squares

Two statements look alike:

- $E[W_T^2] = T$: the **average** of W_T^2 over many paths is T;
- $[W]_T = T$: on **each** path, the sum of squared moves is T.

They are very different. Five simulated paths, each cut into 65,536 steps:

W_1^2 varies wildly from path to path; the quadratic variation is 1 on every path.

PATH	W_1^2	$\Sigma(\Delta W)^2$
1	0.0008	1.0023
2	2.6565	1.0005
3	1.6369	0.9933
4	0.4764	0.9977
5	1.3398	0.9978

W_1^2 is anything from nearly 0 to more than 2: only its average is 1. The quadratic variation is 1 on every path. The first is a statement about the distribution of one number; the second is a property of the path's roughness, which is the same on all paths.

That is why Itô calculus can replace $(dW)^2$ by dt **inside a calculation on one path**, not just on average: Lesson 16 relies on exactly this.

What to remember

- The slope of a Brownian path over a step h is $N(0, 1/h)$: it blows up. Brownian paths have **no derivative** anywhere.
- **First variation** $\sum |\Delta W| \approx \sqrt{2nT/\pi} \to \infty$: the path is infinitely long.
- **Quadratic variation** $\sum (\Delta W)^2 \to T$, with variance $2T^2/n$: $[W]_T = T$, on every path. Smooth paths have quadratic variation 0.
- For the log price, $\sum (\Delta \ln S)^2 \to \sigma^2 T$: realised variance reveals σ.
- **Multiplication table:** $dt\,dt = 0$, $dW\,dt = 0$, $dW\,dW = dt$; correlated Brownian motions: $dW\,dB = \rho\,dt$.
- $E[W_T^2] = T$ is an average over paths; $[W]_T = T$ holds path by path.

Common mistakes

- **"No derivative means the path jumps."** Brownian paths are continuous. They just wiggle too much, at every scale, for a slope to exist.
- **"$\sum (\Delta W)^2$ goes to 0, like for smooth paths."** Each square is of size Δt, not Δt^2, and there are $T/\Delta t$ of them.
- **"$dW\,dW = dt$ means $(W_{t+h} - W_t)^2 = h$ exactly."** For one step it is h times a random Z^2. It is the **sum** over many steps that becomes non-random.
- **Confusing $E[W_T^2] = T$ with $[W]_T = T$.** One is an average, the other a path property.
- **"$dW\,dt$ is of order dt."** It is of order $\Delta t^{3/2}$ and sums to 0.
- **Forgetting the volatility.** For $X = \sigma W$, $(dX)^2 = \sigma^2\,dt$, not dt.

Exercises

Solutions are in *Part 3: Solutions*.

14.1 The slope of a Brownian path over a step h is $Q_h = (W_{t+h} - W_t)/h$. Find its distribution, and compute $P(|Q_h| > 1)$ for $h = 0.01$ and for $h = 0.0001$.

14.2 Show that $E\sum_{i=1}^{n} |\Delta W_i| = \sqrt{2nT/\pi}$ when $[0, T]$ is cut into n equal steps. Evaluate it for $T = 1$ and $n = 100$, 10,000 and 1,000,000.

14.3 Show that $Q_n = \sum_{i=1}^{n} (\Delta W_i)^2$ has mean T and variance $2T^2/n$ (use $E[Z^4] = 3$), and use Chebyshev's inequality to show $Q_n \to T$.

14.4 Show that $\sum \Delta W_i\,\Delta t$ has mean 0 and variance $T\Delta t^2$, and that $\sum \Delta t^2 = T\Delta t$. Why do both go to 0, while $\sum \Delta W_i^2$ does not?

14.5 Find the quadratic variation over $[0, T]$ of (a) the smooth path $f(t) = 100e^{0.03t}$, (b) σW_t, (c) $\ln S_t$ with $S_t = S_0 e^{(r - \frac{1}{2}\sigma^2)t + \sigma W_t}$.

14.6 Let $B = \rho W + \sqrt{1 - \rho^2}\, W'$ with W' a Brownian motion independent of W. Show that B is a Brownian motion and that $\sum \Delta W_i \, \Delta B_i \to \rho T$.

14.7 *(Code)* Simulate one year of daily log prices (252 steps) with $\sigma = 20\%$, and compute the **realised variance** $\sum (\Delta \ln S)^2$. Repeat 10,000 times: what are its mean and standard deviation, and how do they compare with $\sigma^2 T$?

14.8 *(Code)* On one path cut into 2^k steps for $k = 4, \dots, 20$, compute the first variation and the quadratic variation, and describe how each behaves as the number of steps grows.

Self-test

1. Why does $(W_{t+h} - W_t)/h$ not settle as $h \to 0$?
2. What is the quadratic variation of $\ln S$ over one year with $\sigma = 20\%$?
3. On a single path, $W_1^2 = 0.3$. What is that path's quadratic variation over $[0, 1]$?

Companion notebook: `s5e14_brownian_paths.ipynb`. Every number in this chapter is computed by it.

LESSON 15 · VIDEO 15

The Itô Integral

Why trading gains need a new kind of integral; the construction from left-endpoint sums, and why left, right and middle give different answers for a Brownian path; the three properties (martingale, Itô isometry, quadratic variation), each explained; the integral of W dW, worked from a telescoping sum; and why an ordinary function integrated against dW gives a normal random variable.

Why this lesson

In Lesson 10 a hedger held Δ_n shares from step n to step $n+1$ and gained $\Delta_n(S_{n+1} - S_n)$. Over the whole tree the gain was $\sum_n \Delta_n(S_{n+1} - S_n)$. In continuous time the steps shrink to nothing, and the sum should become an integral: $\int_0^T \Delta_t \, dS_t$. Every self-financing portfolio in Part 4 is written this way, and the Black-Scholes argument is a statement about such integrals.

But Lesson 14 showed that a Brownian path is infinitely long and has no slope. The integrals of ordinary calculus need one or the other. So we need a new definition: the **Itô integral** $\int_0^T H_t \, dW_t$. This lesson builds it, shows why the "left endpoint" in its definition is essential (it is the hedger's rule: decide first, then see the move), and works out its three key properties.

What you need from earlier lessons

- **Brownian motion and quadratic variation** (Lessons 13 and 14). Independent increments $\Delta W_i \sim N(0, \Delta t)$; $\sum (\Delta W_i)^2 \to T$; $dW \, dW = dt$.
- **Adapted processes** (Lesson 7). H_t is adapted if its value is known at time t: it can be computed from the path up to t.
- **Martingales and the tower rule** (Lessons 8 and 11).
- **Normal distributions** (Lesson 4). Independent normals add up to a normal, with variances adding.

Notation. As in Lesson 14: n steps of length Δt on $[0, T]$, times t_i, moves $\Delta W_i = W_{t_{i+1}} - W_{t_i}$.

The picture

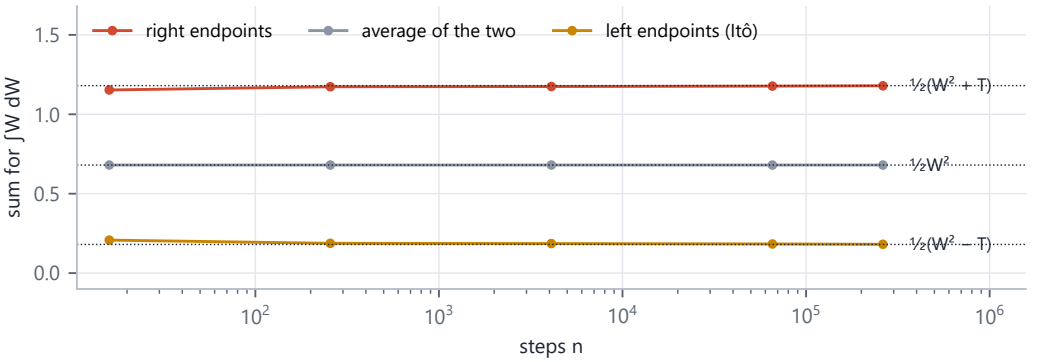


Figure 15.1. Sums for $\int_0^1 W\,dW$ on one Brownian path, evaluating W at the left end of each step (amber), the right end (red), or the average of the two (grey), as the steps shrink. They settle on three different numbers: ½(W² − T), ½(W² + T) and ½W². The gap between left and right is the quadratic variation, T.

For a smooth path all three sums would settle on the same number. For a Brownian path they don't, and the gap between them is exactly the quadratic variation of Lesson 14. We have to choose, and the choice is not arbitrary.

Gains from trading: why a new integral

Riemann sums need short, tame paths

Lesson 1 defined $\int_0^T f(t)\,dt$ as the limit of sums $\sum f(s_i)\Delta t$, where s_i is any point in the i-th step: for a continuous f the choice doesn't matter in the limit. For $\int_0^T H\,dX$ against another path X, the analogous sums are $\sum H(s_i)\Delta X_i$. When X is smooth (or at least has finite length), the choice of s_i doesn't matter either: changing H by a small amount on each step changes the sum by at most (small amount) $\times \sum |\Delta X_i|$, and $\sum |\Delta X_i|$ is finite.

For Brownian motion $\sum |\Delta W_i| \to \infty$ (Lesson 14). The argument fails, and, as the picture shows, so does the conclusion: where you evaluate H in each step changes the answer.

The trader's rule decides

Think of H_t as the number of shares held and W as the price. At t_i you choose H_{t_i}, using what you know then; over the step the price moves by ΔW_i; your gain is $H_{t_i}\Delta W_i$. You cannot use $W_{t_{i+1}}$ to decide how much to hold during the step: that would be trading on tomorrow's price.

Definition. For an adapted H (with $E\int_0^T H_t^2\,dt < \infty$), the **Itô integral** is the limit of left-endpoint sums:

$$\int_0^T H_t\,dW_t = \lim_{n\to\infty} \sum_{i=0}^{n-1} H_{t_i}\,(W_{t_{i+1}} - W_{t_i}).$$

The limit is taken in mean square: the expected squared difference between the sum and the integral goes to 0. (The isometry below is what makes the limit exist.)

An example. Hold one share whenever the path is above where it started, none otherwise: $H_t = 1$ if $W_t > 0$, else 0. On the notebook's path, which ends at $W_1 = 1.1665$ and spends 76% of its time above 0, the gain $\int_0^1 H\,dW$ is 0.3957. Simple to compute, but there is no formula from ordinary calculus for it.

Left, right or middle: it matters

Take $H = W$ itself: $\int_0^T W\,dW$. Three candidate sums:

- **left**: $L_n = \sum W_{t_i}\,\Delta W_i$;
- **right**: $R_n = \sum W_{t_{i+1}}\,\Delta W_i$;
- **average**: $M_n = \sum \tfrac{1}{2}(W_{t_i} + W_{t_{i+1}})\,\Delta W_i$.

Step 1. Right minus left. $R_n - L_n = \sum (W_{t_{i+1}} - W_{t_i})\Delta W_i = \sum (\Delta W_i)^2 \to T$. The gap is the quadratic variation. For a smooth path it would tend to 0; here it doesn't.

Step 2. The average telescopes. $\tfrac{1}{2}(W_{t_i} + W_{t_{i+1}})(W_{t_{i+1}} - W_{t_i}) = \tfrac{1}{2}(W_{t_{i+1}}^2 - W_{t_i}^2)$, a difference of squares. Adding over i, everything cancels except the ends:

$$M_n = \tfrac{1}{2}W_T^2 \quad \text{exactly, for every } n.$$

Step 3. The left sum. $L_n = M_n - \tfrac{1}{2}(R_n - L_n) = \tfrac{1}{2}W_T^2 - \tfrac{1}{2}\sum (\Delta W_i)^2$, so

$$L_n \to \tfrac{1}{2}W_T^2 - \tfrac{1}{2}T, \qquad R_n \to \tfrac{1}{2}W_T^2 + \tfrac{1}{2}T.$$

On the notebook's path, $W_1 = 1.1665$:

The three sums for $\int_0^1 W\,dW$ on one path. Limits: ½(W² − 1), ½(W² + 1) and ½W².

STEPS N	LEFT	RIGHT	AVERAGE	RIGHT − LEFT
16	0.2077	1.1531	0.6804	0.9454
256	0.1874	1.1734	0.6804	0.9859
4,096	0.1860	1.1748	0.6804	0.9888
65,536	0.1834	1.1775	0.6804	0.9941
262,144	0.1814	1.1794	0.6804	0.9979

The limits are 0.1804, 1.1804 and 0.6804.

The left-endpoint integral is the **Itô integral**; the average version is the **Stratonovich integral**, which keeps the rules of ordinary calculus ($\int W\,dW = \frac{1}{2}W^2$) but peeks into the future. Finance uses Itô, because a strategy cannot peek. The price of that choice is the extra $-\frac{1}{2}T$, and it is the same extra term that will turn up in Itô's lemma.

Three properties

Let $I_t = \int_0^t H_s\,dW_s$ with H adapted.

1. It is a martingale

Why, for one step. The step's contribution is $H_{t_i}\Delta W_i$. Given $\mathcal{F}_{t_i}$, H_{t_i} is known (take it out) and ΔW_i is independent with mean 0:

$$E[H_{t_i}\Delta W_i \mid \mathcal{F}_{t_i}] = H_{t_i}\,E[\Delta W_i] = 0.$$

Each step adds nothing on average, given the past, so the sums are martingales, and so is their limit. In particular $E[I_T] = 0$.

In trading terms: if the price is a martingale (a fair game), no strategy that decides before each move can make money on average. This is Lesson 11's "no strategy beats a martingale", in continuous time.

(With right endpoints, $W_{t_{i+1}}$ is *not* known at t_i and the argument breaks: $E[R_n] = E[L_n] + T = T$, not 0.)

2. The Itô isometry

$$E\Big[\Big(\int_0^T H_t\,dW_t\Big)^2\Big] = E\int_0^T H_t^2\,dt.$$

Why. Square the sum: $\big(\sum_i H_{t_i}\Delta W_i\big)^2 = \sum_i H_{t_i}^2(\Delta W_i)^2 + 2\sum_{i<j} H_{t_i}\Delta W_i\,H_{t_j}\Delta W_j$.

- **Squares.** Given $\mathcal{F}_{t_i}$, $E[H_{t_i}^2(\Delta W_i)^2 \mid \mathcal{F}_{t_i}] = H_{t_i}^2\Delta t$. Averaging, $E\sum H_{t_i}^2(\Delta W_i)^2 = E\sum H_{t_i}^2\Delta t \to E\int H^2\,dt$.
- **Cross terms** ($i < j$). Everything except ΔW_j is known at t_j, and $E[\Delta W_j] = 0$. So each cross term averages to 0.

Meaning. The variance of a trading gain equals the average of the squared position, accumulated over time. "Isometry" (equal measure) because it says the size of the integral, measured by its mean square, equals the size of the integrand, measured by $E\int H^2\,dt$.

3. Quadratic variation

Path by path, $[I]_T = \int_0^T H_t^2\,dt$. Each step's move is $H_{t_i}\Delta W_i$, whose square is $H_{t_i}^2(\Delta W_i)^2$, and by Lesson 14 these squares add up like $H_{t_i}^2\,\Delta t$. In differentials: $(dI)^2 = H^2(dW)^2 = H^2\,dt$.

Checked on 20,000 paths, for H = W

$E\int_0^1 W_t^2\,dt = \int_0^1 t\,dt = \frac{1}{2}$, so the isometry says $E[I_1^2] = \frac{1}{2}$.

	SIMULATED	THEORY
$E[I_1]$	0.0051	0
$E[I_1^2]$	0.5098	0.5
$E\int_0^1 W^2\,dt$	0.5004	0.5
average gap between $[I]_1$ and $\int W^2\,dt$	0.0243	0 as steps shrink

(With 1000 steps, the left sums are still a little away from their limit; the gaps shrink as the steps do.)

The integral of W dW

The left-endpoint calculation above gives the formula:

$$\boxed{\int_0^T W_t\,dW_t = \tfrac{1}{2}W_T^2 - \tfrac{1}{2}T}$$

Check with the properties.

- *Mean:* $E[\frac{1}{2}W_T^2 - \frac{1}{2}T] = \frac{1}{2}T - \frac{1}{2}T = 0$. ✓ (martingale)
- *Variance:* $\mathrm{Var}(\frac{1}{2}W_T^2) = \frac{1}{4}(E[W_T^4] - T^2) = \frac{1}{4}(3T^2 - T^2) = \frac{1}{2}T^2$, and the isometry gives $E\int_0^T W^2\,dt = \int_0^T t\,dt = \frac{1}{2}T^2$. ✓

Its distribution. $\frac{1}{2}(W_1^2 - 1)$ is never below $-\frac{1}{2}$ and is negative exactly when $|W_1| < 1$, which happens with probability $2N(1) - 1 = 0.6827$. The simulation gives 0.6794, and a variance of 0.5098 (theory 0.5). So the "gain" from holding W shares is usually a small loss, occasionally a large gain, and zero on average: it is a squared normal, shifted down.

Ordinary calculus would say $\frac{1}{2}W_T^2$. The extra $-\frac{1}{2}T$ is what you lose by having to decide each holding before the move. Lesson 16 shows it is the simplest case of Itô's lemma.

A deterministic integrand gives a normal

If the integrand is an ordinary function of time, $H_t = f(t)$ (known in advance, so certainly adapted), the integral has a simple distribution.

Claim. $\int_0^T f(t)\,dW_t \sim N\Big(0, \int_0^T f(t)^2\,dt\Big)$.

Why. The sum $\sum f(t_i)\Delta W_i$ is a sum of **independent normals** $f(t_i)\Delta W_i \sim N(0, f(t_i)^2\Delta t)$. A sum of independent normals is normal (Lesson 4), with mean 0 and variance $\sum f(t_i)^2\Delta t \to \int_0^T f^2\,dt$. Limits of normals are normal.

Example. $J = \int_0^1 t\,dW_t$ has variance $\int_0^1 t^2\,dt = \frac{1}{3}$.

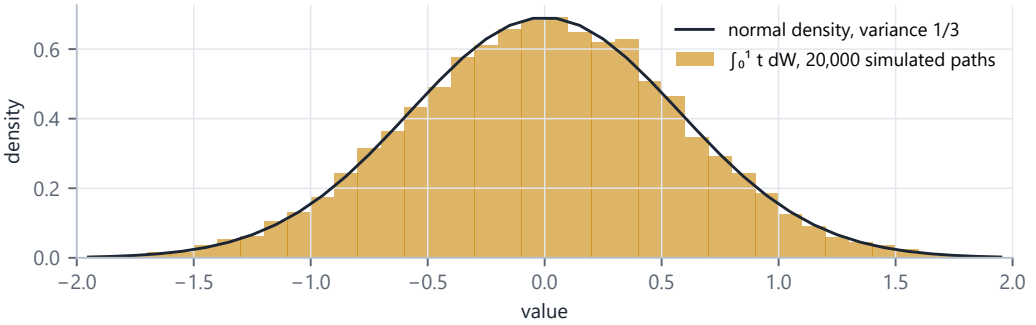


Figure 15.2. 20,000 simulated values of ∫₀¹ t dW against the normal density with mean 0 and variance 1/3.

Simulated: mean 0.0008, variance 0.3372 (theory 0.3333). And $P(J > 0.5) = 1 - N(0.5\sqrt{3}) = 0.1932$, simulated 0.1938.

Covariance with W_1. $W_1 = \int_0^1 1\,dW_t$, and for two integrals of ordinary functions, $E\big[\int f\,dW \int g\,dW\big] = \int fg\,dt$ (the isometry applied to $f + g$). So $\mathrm{Cov}(W_1, J) = \int_0^1 t\,dt = 0.5$; simulated 0.5068.

This fact is used constantly: interest-rate models such as Vasicek (Part 5) produce rates of the form "a function of time plus $\int f(s)\,dW_s$", which are therefore normal.

What to remember

- Trading gains are $\int H\,dW$, with H chosen at the **start** of each step: the **Itô integral** is the limit of left-endpoint sums, for adapted H.
- Left, right and middle sums differ for Brownian paths; right minus left $\to$ the quadratic variation T.
- **Martingale:** $E[\int_0^T H\,dW] = 0$, and $\int_0^t H\,dW$ is a martingale.
- **Isometry:** $E[(\int H\,dW)^2] = E\int H^2\,dt$.

- **Quadratic variation:** $[\int H\,dW]_T = \int H^2\,dt$.
- $\int_0^T W\,dW = \frac{1}{2}W_T^2 - \frac{1}{2}T$.
- $\int_0^T f(t)\,dW_t \sim N(0, \int f^2\,dt)$ for an ordinary function f.

Common mistakes

- **Using ordinary calculus:** $\int W\,dW = \frac{1}{2}W^2$ is the Stratonovich answer, not the Itô one.
- **Evaluating the integrand at the right end.** It is not adapted, the result is not a martingale, and no trader could earn it.
- **Forgetting the square in the isometry.** $E[(\int H\,dW)^2] = E\int H^2\,dt$, not $(E\int H\,dt)^2$.
- "$E[\int H\,dW] = \int E[H]\,dW$." The mean is simply 0 (for adapted H); the right side doesn't make sense as a number.
- "$\int f(t)\,dW$ **has variance** $(\int f\,dt)^2$." It is $\int f^2\,dt$.

Exercises

Solutions are in *Part 3: Solutions*.

15.1 Show that $\sum_i W_{t_i}\Delta W_i = \frac{1}{2}W_T^2 - \frac{1}{2}\sum_i(\Delta W_i)^2$ exactly, for any partition, and deduce the limit.

15.2 Find the limit of the right-endpoint sum $\sum_i W_{t_{i+1}}\Delta W_i$. Why can't a trader earn the right-endpoint sum?

15.3 Use the martingale property and the isometry to find the mean and variance of $\int_0^T W_t\,dW_t$. Check against $\frac{1}{2}(W_T^2 - T)$ directly, using $E[W_T^4] = 3T^2$.

15.4 Show that $J = \int_0^1 t\,dW_t$ is normal with mean 0 and variance $\frac{1}{3}$, and compute $P(J > 0.5)$.

15.5 Compute $\operatorname{Cov}\big(W_1, \int_0^1 t\,dW_t\big)$. (Write $W_1 = \int_0^1 1\,dW_t$; for two integrals, $E[\int f\,dW \int g\,dW] = \int fg\,dt$.)

15.6 Show that $\int_0^T t\,dW_t = TW_T - \int_0^T W_t\,dt$, starting from the discrete identity $t_{i+1}W_{t_{i+1}} - t_iW_{t_i} = t_i\Delta W_i + W_{t_{i+1}}\Delta t$.

15.7 *(Code)* On one Brownian path, compute the left, right and average sums for $\int_0^1 W\,dW$ with 2^k steps, $k = 4, \dots, 18$, and compare with $\frac{1}{2}(W_1^2 \mp 1)$ and $\frac{1}{2}W_1^2$.

15.8 *(Code)* Simulate 100,000 values of $\int_0^1 t\,dW_t$ and of $\int_0^1 W_t\,dW_t$. Compare means and variances with the theory, and the probability that each is negative.

Self-test

1. Why does the Itô integral use the left end of each step?
2. What are the mean and variance of $\int_0^1 W_t\,dW_t$?
3. What is the distribution of $\int_0^2 3\,dW_t$?

Companion notebook: `s5e15_ito_integral.ipynb`. Every number in this chapter is computed by it.

LESSON 16 · VIDEO 16

Itô's Lemma

Itô processes, the continuous-time model of a price; where the extra term in Itô's lemma comes from, from Taylor's theorem and dW·dW = dt, with a picture; the lemma checked on a single simulated path; the special case of a function of W; the general form, and how the input's volatility moves the output's drift; and a moment computed with no density at all, $E[W^4] = 3T^2$.

Why this lesson

The chain rule of ordinary calculus says that if x changes by dx, then $f(x)$ changes by $f'(x)\,dx$. Every derivation in finance needs the equivalent for random paths: if the share moves, how does an option on it move? How does $\ln S$ move? How does S^2 move?

Itô's lemma is that chain rule. It has one extra term, $\frac{1}{2}f''(x)\,(dx)^2$, which ordinary calculus throws away because it is too small. For a random path it isn't too small: by Lesson 14, $(dW)^2 = dt$. Everything in Parts 4 and 5 depends on that term:

- the $-\frac{1}{2}\sigma^2$ in the share's log return (Lesson 17);
- the $\frac{1}{2}\sigma^2 S^2 V_{SS}$ in the Black-Scholes equation (Lesson 19), the term that pays for gamma;
- the drift correction in Girsanov's theorem (Lesson 25).

What you need from earlier lessons

- **Taylor's theorem** (Lessons 1 and 2). $f(x+h) - f(x) = f'(x)h + \frac{1}{2}f''(x)h^2 + O(h^3)$; with two variables, $\Delta f \approx f_t\Delta t + f_x\Delta x + \frac{1}{2}f_{xx}\Delta x^2 + f_{tx}\Delta t\Delta x + \frac{1}{2}f_{tt}\Delta t^2$.
- **The multiplication table** (Lesson 14). $dt\,dt = 0$, $dW\,dt = 0$, $dW\,dW = dt$.
- **The Itô integral** (Lesson 15). $\int H\,dW$ is a martingale with mean 0; $\int_0^T W\,dW = \frac{1}{2}W_T^2 - \frac{1}{2}T$.
- **Differential equations** (Lesson 2). $m' = km$ gives $m(t) = m(0)e^{kt}$.

The picture

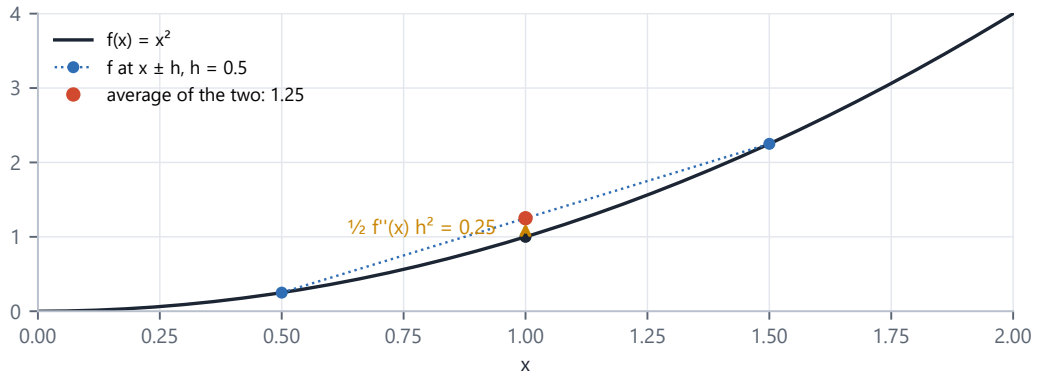


Figure 16.1. For a convex f (here f(x) = x²), a symmetric move ±h from x raises the average of f, from f(x) to the midpoint of the dotted chord. The rise is ½f''(x)h². For a random step, h² is of order dt, and this rise becomes Itô's correction.

Move $\pm h$ with equal chances. On average x doesn't move, but $f(x)$ does: it rises, because f bends upwards. That rise is $\frac{1}{2}f''(x)h^2$, the second term of Taylor's theorem. For a smooth path, h would be of order dt and h^2 negligible. For a Brownian path, h is of order $\sqrt{dt}$ and $h^2 = dt$: the rise is a drift.

Itô processes

Definition. An **Itô process** is a process of the form

$$X_t = X_0 + \int_0^t a_s\, ds + \int_0^t b_s\, dW_s,$$

with adapted a (the **drift**) and b (the **volatility** or **diffusion coefficient**). We write it in differential form:

$$dX_t = a_t\, dt + b_t\, dW_t.$$

The differential form is shorthand for the integral one; it has no meaning on its own (a Brownian path has no derivative, Lesson 14).

Reading it. Over a short step Δt: X moves by $a\,\Delta t$ (predictable) plus $b\,\Delta W$ (random, of size $b\sqrt{\Delta t}$). So the mean of the move is $a\Delta t$ and its variance is $b^2\Delta t$.

Example. $a = 0.03$, $b = 0.2$, constant: $X_t = 0.03t + 0.2W_t$, a Brownian motion with drift. At $t = 1$ it has mean 0.03 and variance 0.04; simulated, 0.0302 and 0.0399.

The squared move. By the multiplication table,

$$(dX)^2 = (a\,dt + b\,dW)^2 = a^2(dt)^2 + 2ab\,dt\,dW + b^2(dW)^2 = b^2\,dt.$$

Only the volatility survives. Keep this: it is half of Itô's lemma.

Where the extra term comes from

The derivation for f(W)

Let f be a smooth function. Cut $[0, T]$ into n steps.

Step 1. Telescope. The total change is the sum of the changes over the steps:

$$f(W_T) - f(W_0) = \sum_i \big(f(W_{t_{i+1}}) - f(W_{t_i})\big).$$

Step 2. Taylor on each step, with $h = \Delta W_i$:

$$f(W_{t_{i+1}}) - f(W_{t_i}) = f'(W_{t_i})\Delta W_i + \tfrac{1}{2}f''(W_{t_i})(\Delta W_i)^2 + (\text{terms of order } |\Delta W_i|^3).$$

Step 3. Add up, term by term.

- $\sum f'(W_{t_i})\Delta W_i$ is a left-endpoint sum: it tends to the Itô integral $\int_0^T f'(W_t)\,dW_t$.
- $\sum \frac{1}{2}f''(W_{t_i})(\Delta W_i)^2$: by Lesson 14, the squares $(\Delta W_i)^2$ add up like Δt, so this tends to $\int_0^T \frac{1}{2}f''(W_t)\,dt$. **This is the term ordinary calculus loses.**
- The cubic terms: each is of order $\Delta t^{3/2}$, and there are $n = T/\Delta t$ of them, so their total is of order $\sqrt{\Delta t} \to 0$.

Step 4. The result.

$$f(W_T) - f(W_0) = \int_0^T f'(W_t)\,dW_t + \int_0^T \tfrac{1}{2}f''(W_t)\,dt,$$

or in differential form, **Itô's lemma for $f(W)$**:

$$\boxed{df(W_t) = f'(W_t)\,dW_t + \tfrac{1}{2}f''(W_t)\,dt}$$

A quick way to remember it

Write Taylor to second order and apply the multiplication table:

$$df = f'\,dW + \tfrac{1}{2}f''\,(dW)^2 = f'\,dW + \tfrac{1}{2}f''\,dt.$$

Itô's lemma on one path

The derivation says the formula holds **path by path**, not just on average. The notebook checks it on one path, which ends at $W_1 = 0.1161$, for $f(x) = x^2$ and $f(x) = e^x$.

For $f(x) = x^2$ the change is $W_1^2 - 0 = 0.0135$:

STEPS	FIRST-ORDER SUM $\sum f'\Delta W$	GAP LEFT	WITH $\sum \frac{1}{2}f''\Delta t$ ADDED	GAP LEFT
64	-1.0637	1.0772	-0.0637	0.0772
1,024	-0.9494	0.9629	0.0506	-0.0371
16,384	-0.9805	0.9940	0.0195	-0.0060
262,144	-0.9900	1.0035	0.0100	0.0035

The first-order sum misses by about 1, whatever the number of steps: that is $\sum \frac{1}{2} \cdot 2 \cdot \Delta t = T$. With Itô's term added, the gap shrinks to 0. For $f(x) = e^x$ the change is 0.1231; with the finest grid the first-order sum misses by 0.8515, and Itô's version by 0.0025.

Functions of W

Apply the boxed formula:

$f(x)$	f'	f''	$df(W)$
x^2	$2x$	2	$2W\,dW + dt$
x^3	$3x^2$	$6x$	$3W^2\,dW + 3W\,dt$
e^x	e^x	e^x	$e^W\,dW + \frac{1}{2}e^W\,dt$

The first row is Lesson 15's integral. Integrating, $W_T^2 = 2\int_0^T W\,dW + T$, so $\int_0^T W\,dW = \frac{1}{2}W_T^2 - \frac{1}{2}T$. ✓

Taking averages. The dW parts are Itô integrals, with mean 0 (Lesson 15). So averages see only the dt parts. For $m(t) = E[e^{W_t}]$, the third row gives

$$m(t) - 1 = E\int_0^t \tfrac{1}{2}e^{W_s}\,ds = \int_0^t \tfrac{1}{2}m(s)\,ds, \qquad \text{so} \qquad m'(t) = \tfrac{1}{2}m(t), \quad m(t) = e^{t/2}.$$

That is the lognormal mean $E[e^{W_t}] = e^{t/2}$ from Lesson 4, found without integrating a density. Simulated: 1.2836, 1.6487, 2.7206 at $t = 0.5, 1, 2$, against $e^{t/2} =$ 1.2840, 1.6487, 2.7183.

The general form: the input's volatility moves the output's drift

Now let X be an Itô process, $dX = a\,dt + b\,dW$, and let $f(t, x)$ depend on time as well. Taylor in two variables, then the multiplication table:

$$df = f_t\,dt + f_x\,dX + \tfrac{1}{2}f_{xx}(dX)^2 + f_{tx}\,dt\,dX + \tfrac{1}{2}f_{tt}(dt)^2.$$

Step 1. $dt\,dX = a(dt)^2 + b\,dt\,dW = 0$ and $(dt)^2 = 0$: the last two terms vanish.

Step 2. $(dX)^2 = b^2\,dt$ (from the section on Itô processes).

Step 3. Substitute $dX = a\,dt + b\,dW$ and collect the dt and dW terms:

$$\boxed{df(t, X_t) = \left(f_t + a\,f_x + \tfrac{1}{2}b^2 f_{xx}\right)dt + b\,f_x\,dW_t}$$

Read it: input and output. The **input** process X has drift a and volatility b. The **output** process $f(t, X)$ has

- volatility $b\,f_x$: the input's volatility, scaled by the slope (as in ordinary calculus);
- drift $f_t + a\,f_x + \tfrac{1}{2}b^2 f_{xx}$: the ordinary chain-rule terms, **plus** a term that depends on the input's *volatility* and the output's *curvature*.

So a purely random input (no drift) can produce an output with drift, if f is curved. That is the convexity picture at the top of the lesson.

Example: from log price to price. Let the log price be $X_t = (r - \tfrac{1}{2}\sigma^2)t + \sigma W_t$: drift $a = r - \tfrac{1}{2}\sigma^2 = 0.0100$, volatility $b = \sigma$. The price is $S = S_0 e^X$, with $f_x = f_{xx} = S$, $f_t = 0$:

$$dS = \left((r - \tfrac{1}{2}\sigma^2)S + \tfrac{1}{2}\sigma^2 S\right)dt + \sigma S\,dW = rS\,dt + \sigma S\,dW.$$

The log grows at 0.01 a year; the price at 0.03; the difference, $\tfrac{1}{2}\sigma^2 = 0.02$, is Itô's correction. Simulated on 50,000 paths:

The input (log price) and the output (price) grow at different rates.

TIME	AVERAGE LOG RETURN	(r − σ²/2)t	AVERAGE PRICE	S₀e^(rt)
0.25	0.0026	0.0025	100.76	100.75
0.50	0.0045	0.0050	101.46	101.51
1.00	0.0091	0.0100	102.95	103.05

Lesson 17 runs this the other way: start from $dS = \mu S\,dt + \sigma S\,dW$ and use Itô on $\ln S$ to solve it.

A moment by Itô: E[W⁴] = 3T²

Lesson 4 computed $E[Z^4] = 3$ by integrating against the normal density. Itô gives it with no integral of a density at all.

Step 1. $f(x) = x^4$: $f' = 4x^3$, $f'' = 12x^2$, so $d(W^4) = 4W^3\,dW + 6W^2\,dt$.

Step 2. Average. The dW term has mean 0, so $E[W_T^4] = \int_0^T 6E[W_t^2]\,dt = \int_0^T 6t\,dt = 3T^2$.

E[W_T⁴] by simulation and by Itô.

T	Simulated E[W⁴]	3T²
0.5	0.7476	0.7500
1.0	2.9903	3.0000
2.0	11.9611	12.0000

The same method gives every moment from the one before (Exercise 16.4 does $E[W^6] = 15T^3$).

What to remember

- **Itô process:** $dX = a\,dt + b\,dW$; then $(dX)^2 = b^2\,dt$.
- **Itô's lemma for $f(W)$:** $df(W) = f'(W)\,dW + \frac{1}{2}f''(W)\,dt$.
- **General form:** $df(t,X) = (f_t + af_x + \frac{1}{2}b^2 f_{xx})\,dt + bf_x\,dW$.
- **Recipe:** Taylor to second order; replace $(dW)^2$ by dt and drop dt^2 and $dt\,dW$.
- The correction $\frac{1}{2}b^2 f_{xx}$ is the **input's volatility** moving the **output's drift**, through the curvature of f.
- It holds path by path, not just on average.
- **Averages:** the dW terms have mean 0, so $\frac{d}{dt}E[f(W_t)] = E[\frac{1}{2}f''(W_t)]$; e.g. $E[e^{W_t}] = e^{t/2}$, $E[W_t^4] = 3t^2$.

Common mistakes

- **Dropping the second-order term.** That is ordinary calculus, and it is wrong by a drift for any curved f.
- **Using $(dX)^2 = dt$.** It is $b^2\,dt$: the volatility squared.
- **Keeping $dt\,dW$ or $(dt)^2$.** They add up to 0.
- **Forgetting f_t** when f depends on time (as for $e^{\sigma W - \frac{1}{2}\sigma^2 t}$).
- **Mixing up a and the output's drift.** The output's drift is $f_t + af_x + \frac{1}{2}b^2 f_{xx}$, not a.
- **Thinking the correction needs a drift.** It comes from the volatility alone; $X = W$, with no drift, still produces $\frac{1}{2}f''\,dt$.

Exercises

Solutions are in *Part 3: Solutions*.

16.1 Use Itô's lemma to find $d(W_t^3)$ and $d(e^{W_t})$. Check each against $f(W_T) - f(0) = \int f'\,dW + \int \frac{1}{2} f''\,dt$ by taking expectations.

16.2 Use the two-variable form to find $d(tW_t)$. Which term would ordinary calculus give, and is there an Itô correction here?

16.3 Show that $Z_t = e^{\sigma W_t - \frac{1}{2}\sigma^2 t}$ satisfies $dZ_t = \sigma Z_t\,dW_t$: it has no dt term.

16.4 Use Itô's lemma on W^6 to show $E[W_T^6] = 15T^3$.

16.5 Let $dX = a\,dt + b\,dW$ with constants a, b and $X_0 = 0$. Find $d(X^2)$, and from it $E[X_T^2]$. Check with $X_T = aT + bW_T$.

16.6 Let $Y_t = \cos W_t$. Find dY_t, take expectations and solve the resulting equation to show $E[\cos W_t] = e^{-t/2}$.

16.7 *(Code)* On one Brownian path, check Itô's lemma for $f(x) = x^3$: compare W_1^3 with $\sum 3W_{t_i}^2 \Delta W_i + \sum 3W_{t_i}\Delta t$ for 2^k steps, $k = 6, \dots, 18$.

16.8 *(Code)* Check $E[\cos W_t] = e^{-t/2}$ and $E[W_t^4] = 3t^2$ by simulation for $t = 0.5, 1, 2$.

Self-test

1. What is $d(W_t^2)$? Where does the dt come from?
2. The log price has volatility $\sigma = 20\%$. By how much does the price's drift exceed the log price's drift?
3. For $dX = a\,dt + b\,dW$, what is $(dX)^2$?

Companion notebook: `s5e16_itos_lemma.ipynb`. Every number in this chapter is computed by it.

LESSON 17 · VIDEO 17

Geometric Brownian Motion

The share-price model dS = μS dt + σS dW, and what each term means; solving it with Itô's lemma on ln S, and where the −½σ² comes from; closing the loop with Lesson 5's lognormal (mean, median, mode); E[S²] and the variance by Itô; and Euler's method against the exact solution, including the error you make by forgetting the −½σ².

Why this lesson

Lesson 5 *assumed* that the share price in a year is lognormal: $S_T = S_0 e^{(r-\frac{1}{2}\sigma^2)T+\sigma\sqrt{T}Z}$. That formula, with its mysterious $-\frac{1}{2}\sigma^2$, came from asking the average price to grow like cash. This lesson **derives** it from a model of how the price moves from moment to moment:

$$dS_t = \mu S_t\,dt + \sigma S_t\,dW_t,$$

geometric Brownian motion (GBM), the model behind Black-Scholes. We solve it with Itô's lemma, recover every Lesson 5 formula, compute the variance a new way, and then do what banks do for models without a formula: simulate the equation step by step, and measure how wrong that is.

What you need from earlier lessons

- **Itô's lemma** (Lesson 16). For $dX = a\,dt + b\,dW$: $df(t,X) = (f_t + af_x + \frac{1}{2}b^2 f_{xx})\,dt + bf_x\,dW$, and $(dX)^2 = b^2\,dt$.
- **The lognormal** (Lesson 5). If $\ln S_T \sim N(m, s^2)$: mean $e^{m+s^2/2}$, median e^m, mode e^{m-s^2}.
- **Differential equations and Euler's method** (Lesson 2). $m' = km \Rightarrow m(t) = m(0)e^{kt}$; Euler steps $m \leftarrow m + km\,\Delta t$.
- **The exponential martingale** (Lesson 13). $e^{\sigma W_t - \frac{1}{2}\sigma^2 t}$ is a martingale.

Numbers. $S_0 = 100$, $\sigma = 20\%$, $T = 1$, and the expected return $\mu = 3\%$, equal to r, so the results can be checked against Lesson 5.

The picture

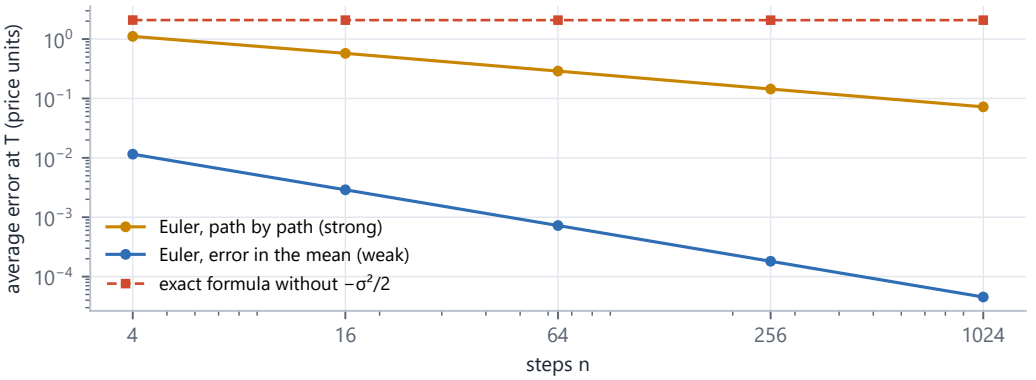


Figure 17.1. Errors at T = 1 when the share's equation is stepped with Euler's method, against the exact solution driven by the same noise. Path by path (amber), the error halves each time the steps are cut by four; in the mean (blue), it quarters. The red line uses the exact formula with the −½σ² deleted: its error never shrinks.

A numerical method that is merely crude gets better with more steps. A formula that is wrong does not. The red line is what happens if you forget Itô's correction.

The model

$$dS_t = \mu S_t\, dt + \sigma S_t\, dW_t.$$

Divide by S_t: over a short step, the **return** $\Delta S/S$ is $\mu\,\Delta t + \sigma\,\Delta W$:

- $\mu\, dt$**:** the expected return, μ per year;
- $\sigma\, dW$**:** a random return with standard deviation $\sigma\sqrt{\Delta t}$.

Over one day ($\Delta t = 1/252$) the return has mean $\mu/252 = 0.000119$ and standard deviation $\sigma/\sqrt{252} = 0.0126$ (simulated: 0.000094 and 0.0126). The random part dominates any single day by a factor of about 100.

Why "geometric". The moves are proportional to the price: a share at 200 moves twice as many dollars as one at 100. Percentages, not dollars, are what stay the same, and the price can never become negative. (A plain Brownian motion with drift, $dS = \mu\, dt + \sigma\, dW$, could.)

Solving it with d ln S

Returns are additive in logs, so try $f(S) = \ln S$.

Step 1. The derivatives. $f' = 1/S$, $f'' = -1/S^2$, $f_t = 0$.

Step 2. Itô's lemma. Here $a = \mu S$ and $b = \sigma S$:

$$\begin{aligned} d\ln S &= \Big(\mu S \cdot \frac{1}{S} + \frac{1}{2}\sigma^2 S^2 \cdot \Big(-\frac{1}{S^2}\Big)\Big)dt + \sigma S \cdot \frac{1}{S}\,dW \\ &= \Big(\mu - \frac{1}{2}\sigma^2\Big)dt + \sigma\,dW. \end{aligned}$$

The S's cancel everywhere: the log price is a Brownian motion with **constant** drift $\mu - \frac{1}{2}\sigma^2$ and volatility σ.

Step 3. Integrate from 0 to t (constants integrate trivially, and $\int_0^t dW = W_t$):

$$\ln S_t - \ln S_0 = \Big(\mu - \frac{1}{2}\sigma^2\Big)t + \sigma W_t.$$

Step 4. Exponentiate.

$$\boxed{S_t = S_0 \exp\Big(\big(\mu - \tfrac{1}{2}\sigma^2\big)t + \sigma W_t\Big)}$$

Where the $-\frac{1}{2}\sigma^2$ comes from. From the $\frac{1}{2}f''(dS)^2$ term: ln is concave ($f'' < 0$), so a symmetric random move **lowers** the average log (the convexity picture of Lesson 16, upside down). The price drifts up at μ; its log drifts up more slowly, at $\mu - \frac{1}{2}\sigma^2$. With our numbers: 0.02 a year slower.

Checked. Over 50,000 simulated years, the average log return is 0.0094 (theory $\mu - \frac{1}{2}\sigma^2 = 0.0100$) and its variance 0.0398 (theory $\sigma^2 = 0.04$).

Check by differentiating (Exercise 17.2): apply the two-variable Itô's lemma to $f(t, w) = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma w}$ and you get back $dS = \mu S\,dt + \sigma S\,dW$.

Back to Lesson 5: the lognormal

$\ln S_T$ is normal with mean $m = \ln S_0 + (\mu - \frac{1}{2}\sigma^2)T$ and variance $s^2 = \sigma^2 T$, so S_T is **lognormal**, and Lesson 5's formulas apply directly:

	FORMULA	VALUE	SIMULATED
Mean	$S_0 e^{\mu T}$	103.0455	102.9666
Median	$S_0 e^{(\mu - \frac{1}{2}\sigma^2)T}$	101.0050	101.0532
Mode	$S_0 e^{(\mu - \frac{3}{2}\sigma^2)T}$	97.0446	
$P(S_T > \text{mean})$	$N(-\frac{1}{2}\sigma\sqrt{T})$	0.4602	0.4617
$P(S_T < S_0)$	$N\big(-(\mu - \frac{1}{2}\sigma^2)\sqrt{T}/\sigma\big)$	0.4801	0.4792

The mean. $E[S_T] = S_0 e^{(\mu - \frac{1}{2}\sigma^2)T} E[e^{\sigma W_T}] = S_0 e^{(\mu - \frac{1}{2}\sigma^2)T} e^{\frac{1}{2}\sigma^2 T} = S_0 e^{\mu T}$: the two $\frac{1}{2}\sigma^2$'s cancel. On average the price grows at exactly μ, as the equation promised.

Less than half the paths beat the average. A few paths end very high and pull the mean up; most end below it. That is why the median is below the mean, and why "the expected price" and "the typical price" are different things.

The discounted price is a martingale when $\mu = r$. $e^{-\mu t} S_t = S_0 e^{\sigma W_t - \frac{1}{2}\sigma^2 t}$, the exponential martingale of Lesson 13. With $\mu = r$, the discounted share is a fair game: this is the risk-neutral world of Part 2, now in continuous time.

E[S²] by Itô

Lesson 5 found $E[S_T^2]$ with the lognormal formula. Itô's lemma gives it without any distribution.

Step 1. Itô on $f(S) = S^2$: $f' = 2S$, $f'' = 2$, so

$$\begin{aligned} d(S^2) &= 2S\,dS + (dS)^2 = 2S(\mu S\,dt + \sigma S\,dW) + \sigma^2 S^2\,dt \\ &= (2\mu + \sigma^2) S^2\,dt + 2\sigma S^2\,dW. \end{aligned}$$

Step 2. Average. The dW term has mean 0, so $m(t) = E[S_t^2]$ satisfies $m'(t) = (2\mu + \sigma^2) m(t)$, with $m(0) = S_0^2$.

Step 3. Solve (Lesson 2): $E[S_T^2] = S_0^2 e^{(2\mu + \sigma^2)T}$. The rate is $2\mu + \sigma^2 = 0.10$, so

$$E[S_1^2] = 10{,}000\, e^{0.10} = 11051.71 \quad \text{(simulated } 11031.05\text{)}.$$

Step 4. Variance. $\mathrm{Var}(S_T) = E[S_T^2] - E[S_T]^2 = S_0^2 e^{2\mu T}(e^{\sigma^2 T} - 1) = 433.34$, a standard deviation of 20.82: Lesson 5's number. Note the σ^2 in the rate: the square is convex, so the $\frac{1}{2} f''(dS)^2$ term adds to the drift this time.

Euler's method against the exact solution

GBM has a formula. Most models don't, and must be simulated step by step. **Euler's method** (often called Euler-Maruyama for random equations) replaces dt by Δt and dW by ΔW:

$$S_{k+1} = S_k\big(1 + \mu\,\Delta t + \sigma\,\Delta W_k\big).$$

How good is it? Drive Euler and the exact formula with the **same** random moves, so the only difference is the method. Two ways to measure the error at T:

- **strong error**, $E|S_T^{\text{Euler}} - S_T^{\text{exact}}|$: how far apart the two are on the same path. It matters when you need the path itself (hedging simulations, path-dependent options).
- **weak error**, $|E[S_T^{\text{Euler}}] - E[S_T^{\text{exact}}]|$: how far apart the averages are. It matters for prices of ordinary options. (Here it can be computed exactly: $E[S_T^{\text{Euler}}] = S_0(1 + \mu T/n)^n$.)

Errors at T = 1 on 20,000 paths (in price units, S_0 = 100).

STEPS N	STRONG ERROR	WEAK ERROR	Formula without −σ²/2
4	1.1104	0.011534	2.0888
16	0.5750	0.002894	2.0845
64	0.2889	0.000724	2.0830
256	0.1441	0.000181	2.0808
1,024	0.0722	0.000045	2.0825

Reading the table.

- **Strong error** halves every time the steps are cut by four (the ratio of the last two rows is 2.00): it shrinks like $\sqrt{\Delta t}$. Euler has **strong order ½**.
- **Weak error** quarters every time: it shrinks like Δt. Euler has **weak order 1**. Averages converge much faster than paths, which is good news for pricing.
- **The deleted $-\frac{1}{2}\sigma^2$.** Using $S_0 e^{\mu T + \sigma W_T}$ is not an approximation that improves: it is the wrong distribution. Its mean is $S_0 e^{(\mu + \frac{1}{2}\sigma^2)T} = 105.13$ instead of 103.05, and its distance from the true path stays at about 2 whatever the number of steps.

Why Euler's strong error is only $\sqrt{\Delta t}$. Expand the exact one-step growth factor to second order: $e^{(\mu - \frac{1}{2}\sigma^2)\Delta t + \sigma \Delta W} \approx 1 + \mu\Delta t + \sigma\Delta W + \frac{1}{2}\sigma^2\big((\Delta W)^2 - \Delta t\big)$. Euler keeps the first three terms and drops the last one, which is of size Δt on each step, random, with mean 0. Adding up n such independent errors gives a total of size $\sqrt{n}\,\Delta t = \sqrt{T\Delta t}$. Keeping that term is the **Milstein scheme** (Exercise 17.8), with strong order 1.

What to remember

- **GBM:** $dS = \mu S\,dt + \sigma S\,dW$; returns have mean $\mu\,dt$ and standard deviation $\sigma\sqrt{dt}$.
- **Solution:** $d\ln S = (\mu - \frac{1}{2}\sigma^2)\,dt + \sigma\,dW$, so $S_t = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}$.
- The $-\frac{1}{2}\sigma^2$ is Itô's correction for the concave ln; the price grows at μ, its log at $\mu - \frac{1}{2}\sigma^2$.
- S_T is lognormal: mean $S_0 e^{\mu T}$, median $S_0 e^{(\mu - \frac{1}{2}\sigma^2)T}$; $P(S_T > E[S_T]) < \frac{1}{2}$.
- $E[S_T^2] = S_0^2 e^{(2\mu + \sigma^2)T}$, by Itô on S^2.

- $e^{-\mu t}S_t$ is a martingale; with $\mu = r$, the discounted price is a fair game.
- **Euler:** strong error $\propto \sqrt{\Delta t}$, weak error $\propto \Delta t$. Deleting $-\frac{1}{2}\sigma^2$ is an error that never shrinks.

Common mistakes

- **Writing $S_t = S_0 e^{\mu t + \sigma W_t}$.** That process grows at $\mu + \frac{1}{2}\sigma^2$, not μ.
- **"The log grows at μ."** It grows at $\mu - \frac{1}{2}\sigma^2$; the price grows at μ.
- **Ordinary calculus on $\ln S$:** $d\ln S = dS/S$ misses $-\frac{1}{2}\sigma^2\,dt$.
- **Judging a simulation by one path.** Strong and weak errors measure different things; know which one your problem needs.
- **Comparing Euler with the exact formula on different random numbers.** Then the difference is dominated by sampling noise, not the method.

Exercises

Solutions are in *Part 3: Solutions*.

17.1 Derive $d\ln S_t$ for $dS = \mu S\,dt + \sigma S\,dW$ with Itô's lemma, step by step, and solve for S_t.

17.2 Check that $S_t = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}$ satisfies the equation, by applying the two-variable Itô's lemma to $f(t, w) = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma w}$.

17.3 With $\mu = 3\%$, $\sigma = 20\%$, $T = 1$, find the mean, median and mode of S_T, and $P(S_T > E[S_T])$. Why is the last one below a half?

17.4 Use Itô's lemma on S^2 to find $E[S_T^2]$ and $\mathrm{Var}(S_T)$. Compare with Lesson 5's formula.

17.5 Show that $e^{-\mu t}S_t$ is a martingale. What does this say when $\mu = r$?

17.6 Find the distribution of $\ln(S_T/S_0)$ and compute $P(S_1 < 100)$ and $P(S_1 < 80)$.

17.7 *(Code)* Compare Euler's method with the exact solution, driven by the same noise, for $n = 4, 16, \ldots, 1024$ steps: the average size of the error at T (strong) and the error in the mean (weak). Estimate how each shrinks.

17.8 *(Code)* The **Milstein** scheme adds one term: $S_{k+1} = S_k\big(1 + \mu\Delta t + \sigma\Delta W_k + \frac{1}{2}\sigma^2(\Delta W_k^2 - \Delta t)\big)$. Repeat the strong-error test and compare with Euler.

Self-test

1. At what rate does $\ln S$ grow on average, with $\mu = 3\%$ and $\sigma = 20\%$?
2. Why is less than half the probability above $E[S_T]$?
3. You halve Euler's time step. Roughly how much smaller do the strong and weak errors get?

Companion notebook: `s5e17_geometric_brownian_motion.ipynb`. Every number in this chapter is computed by it.

LESSON 18 · VIDEO 18

The Itô Toolbox

The quickest test for a martingale (write the process with Itô's lemma and look at the dt term); the exponential martingale, derived in full; polynomial martingales, including a one-line proof that $W^2 - t$ is one; the Itô product rule and its extra dX·dY term, used to show the discounted share is a martingale exactly when μ = r; and a first look at Girsanov's theorem, where weighting paths by an exponential martingale shifts Brownian motion.

Why this lesson

Lessons 13 to 17 built the machinery. This lesson turns it into four tools you will use on almost every page of Part 4:

1. **The martingale test.** Pricing needs martingales (Lesson 11). With Itô's lemma, checking for one takes a line: no dt term, no drift.
2. **The exponential martingale** $e^{\theta W_t - \frac{1}{2}\theta^2 t}$: the continuous-time Radon-Nikodym derivative, and the heart of Girsanov's theorem.
3. **The product rule** for $d(XY)$, needed whenever a price is multiplied by a discount factor or by another price.
4. **Reweighting paths** to shift a Brownian motion: a preview of Lesson 25.

What you need from earlier lessons

- **Itô's lemma** (Lesson 16). $df(t, X) = (f_t + a f_x + \frac{1}{2} b^2 f_{xx})\, dt + b f_x\, dW$ for $dX = a\, dt + b\, dW$; and its two-variable version for $f(t, x, y)$, with the multiplication table $dW\, dW = dt$, $dW\, dB = \rho\, dt$, everything else 0.
- **The Itô integral is a martingale** (Lesson 15), when $E \int H^2\, dt < \infty$.
- **Martingales and the Radon-Nikodym derivative** (Lessons 11 and 12). $\tilde{E}[X] = E[ZX]$.
- **Three martingales** (Lesson 13). W, $W^2 - t$, $e^{\sigma W - \frac{1}{2}\sigma^2 t}$.

The picture

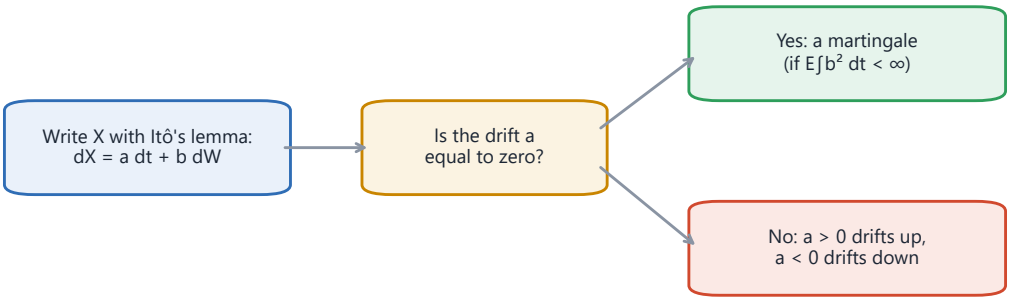


Figure 18.1. The martingale test. Write the process with Itô's lemma; it is a martingale exactly when its dt term (its drift) is zero.

The martingale test

Claim. Let $dX_t = a_t\,dt + b_t\,dW_t$, with $E\int_0^T b_t^2\,dt < \infty$. Then X is a martingale if and only if $a_t = 0$ (for all t, with probability 1).

Why. $X_t = X_0 + \int_0^t a\,ds + \int_0^t b\,dW$. The last piece is an Itô integral, a martingale (Lesson 15). So X is a martingale exactly when $\int_0^t a\,ds$ is one. That integral is a process with no randomness from moment to moment: if it is a fair game, it cannot move, so $a = 0$. (And if $a > 0$, X drifts up on average: a submartingale; if $a < 0$, a supermartingale.)

The condition on b rules out pathological integrands that make the Itô integral misbehave. For everything in this course it holds.

The test in action. Six candidates, written with Itô's lemma:

PROCESS	ITÔ'S LEMMA GIVES	dt TERM
W^3	$3W^2\,dW + 3W\,dt$	$3W$
$W^3 - 3tW$	$3(W^2 - t)\,dW$	0
e^W	$e^W\,dW + \frac{1}{2}e^W\,dt$	$\frac{1}{2}e^W$
$e^{W-t/2}$	$e^{W-t/2}\,dW$	0
$W^4 - 6tW^2 + 3t^2$	$(4W^3 - 12tW)\,dW$	0
tW	$W\,dt + t\,dW$	W

The notebook checks each one the hard way. It simulates 200,000 paths, sorts them into ten groups by their value at time 0.5, and compares the average value at time 1 with the value at time 0.5 in each group. The largest gap over the ten groups:

Where the dt term is zero, the time-1 average matches the time-0.5 value in every group.

PROCESS	DT TERM	LARGEST GAP IN A GROUP	MARTINGALE?
W^3	3W	1.888	no
$W^3 - 3tW$	0	0.038	yes
e^W	½e^W	1.050	no
e^(W − t/2)	0	0.012	yes
$W^4 - 6tW^2 + 3t^2$	0	0.111	yes
tW	W	0.620	no

Note that W^3 has $E[W_t^3] = 0$ for all t, the same as at the start, yet it is **not** a martingale: given $W_{0.5} = 1$, its future average is not 1^3. Constant expectation is necessary, not sufficient. The group test catches the difference; the dt term explains it.

The exponential martingale

Claim. For a constant θ, $Z_t = e^{\theta W_t - \frac{1}{2}\theta^2 t}$ satisfies $dZ_t = \theta Z_t \, dW_t$. So it is a martingale, with $E[Z_t] = 1$.

Proof. $Z_t = f(t, W_t)$ with $f(t, w) = e^{\theta w - \frac{1}{2}\theta^2 t}$.

Step 1. The derivatives. $f_t = -\frac{1}{2}\theta^2 f, \;\; f_w = \theta f, \;\; f_{ww} = \theta^2 f$.

Step 2. Itô's lemma with $a = 0, b = 1$ (the input is W itself):

$$dZ = \Big(-\tfrac{1}{2}\theta^2 Z + 0 + \tfrac{1}{2} \cdot 1 \cdot \theta^2 Z \Big) dt + \theta Z \, dW = \theta Z \, dW .$$

Step 3. Read it. The $-\frac{1}{2}\theta^2$ in the exponent produces a drift of $-\frac{1}{2}\theta^2 Z$; the curvature of the exponential produces $+\frac{1}{2}\theta^2 Z$. They cancel exactly. That is the whole reason for the $-\frac{1}{2}\theta^2 t$.

Without the correction, $e^{\theta W_t}$ has $E[e^{\theta W_1}] = e^{\frac{1}{2}\theta^2}$:

With the correction, the average stays at 1; without it, it grows like e^(θ²/2).

θ	E[Z_1]	E[e^(θW_1)], simulated	e^(θ²/2)
0.2	1.0002	1.0204	1.0202
0.5	1.0005	1.1338	1.1331
1.0	1.0016	1.6514	1.6487
2.0	1.0257	7.5787	7.3891

(For $\theta = 2$ the simulated $E[Z_1]$ is a little off: Z_1 then has a few huge values on rare paths, and 200,000 paths don't see enough of them. A warning for simulations with large weights.)

More general integrands. The same calculation with $X_t = \int_0^t \theta_s\, dW_s - \frac{1}{2}\int_0^t \theta_s^2\, ds$ gives $d(e^X) = \theta_t e^X\, dW$: for adapted θ (and a technical condition, Novikov's, that keeps it from exploding), $e^{\int \theta\, dW - \frac{1}{2}\int \theta^2\, dt}$ is a martingale. Lesson 25 uses this form.

Polynomial martingales

$W^2 - t$, in one line. Lesson 13 needed a paragraph. With Itô:

$$d(W^2 - t) = (2W\, dW + dt) - dt = 2W\, dW.$$

No dt term: a martingale.

A family. The same check works for $W^3 - 3tW$ and $W^4 - 6tW^2 + 3t^2$ (the table above; the middle column is what is left after the dt terms cancel). These are the **Hermite** polynomials in disguise; there is one for every power. Each gives a moment for free:

- $E[W_t^3 - 3tW_t] = 0$ gives $E[W_t^3] = 3tE[W_t] = 0$;
- $E[W_t^4 - 6tW_t^2 + 3t^2] = 0$ gives $E[W_t^4] = 6t \cdot t - 3t^2 = 3t^2$.

Simulated at $t = 1$: $E[W^3 - 3W] = 0.0013$ and $E[W^4 - 6W^2 + 3] = 0.0102$.

The Itô product rule

Claim. For two Itô processes X and Y,

$$\boxed{d(XY) = X\, dY + Y\, dX + dX\, dY}$$

Why. Apply the two-variable Itô's lemma to $f(x, y) = xy$: $f_x = y$, $f_y = x$, $f_{xx} = f_{yy} = 0$, $f_{xy} = 1$. Taylor to second order:

$$d(XY) = Y\, dX + X\, dY + 0 \cdot (dX)^2 + 0 \cdot (dY)^2 + dX\, dY.$$

The extra term. If $dX = a\,dt + b\,dW$ and $dY = c\,dt + e\,dW$ (the same W), then $dX\,dY = be\,dt$. If Y has no dW part (it is an ordinary function of time, like a discount factor), $dX\,dY = 0$ and the rule is the ordinary one.

Example 1: $d(W \cdot W)$. $X = Y = W$: $d(W^2) = W\,dW + W\,dW + dW\,dW = 2W\,dW + dt$. ✓ (Lesson 16.)

Example 2: $d(tW)$. $X = t$ (so $dX = dt$), $Y = W$: $d(tW) = t\,dW + W\,dt + dt\,dW = t\,dW + W\,dt$. So $tW_t - \int_0^t W_s\,ds = \int_0^t s\,dW_s$ is a martingale (Exercise 18.3). The notebook's average of $W_1 - \int_0^1 W\,ds$ is -0.0074 (0), and the variance of $\int_0^1 W\,ds$ is 0.3305 (theory $\frac{1}{3}$, Exercise 15.6).

Example 3: the discounted share. Let $dS = \mu S\,dt + \sigma S\,dW$ and $D_t = e^{-rt}S_t$. Here $X = e^{-rt}$ has $dX = -re^{-rt}\,dt$ and no dW part, so $dX\,dS = 0$:

$$\begin{aligned} dD &= e^{-rt}\,dS + S\,d(e^{-rt}) = e^{-rt}(\mu S\,dt + \sigma S\,dW) - re^{-rt}S\,dt \\ &= (\mu - r)D\,dt + \sigma D\,dW. \end{aligned}$$

By the martingale test, **the discounted share is a martingale exactly when** $\mu = r$: the continuous-time version of Lesson 11's result. Simulated: with $\mu = r$, $E[e^{-r}S_1] =$ 100.02 (theory 100); with $\mu = 8\%$, 105.15 (theory $100e^{0.05} = 105.13$).

In the real world $\mu > r$ (investors want a reward for risk), so the discounted share is a submartingale. Part 4's task is to change the probability so that its drift becomes r, and the next section shows how.

A preview of Girsanov: reweighting shifts the path

In Lesson 12, multiplying the probability of each world by $Z = \tilde{P}/P$ turned the real-world tree into the risk-neutral one. Do the same with Brownian paths, weighting each path by $Z_T = e^{\theta W_T - \frac{1}{2}\theta^2 T}$ (a positive weight with average 1, so it defines a new probability).

Claim. For any function f, $E[Z_T f(W_T)] = E[f(W_T + \theta T)]$.

Why, by completing the square (Lesson 4). With $W_T = \sqrt{T}\,x$ and φ the standard normal density:

$$e^{\theta\sqrt{T}x - \frac{1}{2}\theta^2 T}\varphi(x) = \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(x-\theta\sqrt{T})^2} = \varphi(x - \theta\sqrt{T}).$$

So weighting by Z_T turns the $N(0, T)$ density of W_T into the $N(\theta T, T)$ density: the same as shifting W_T by θT.

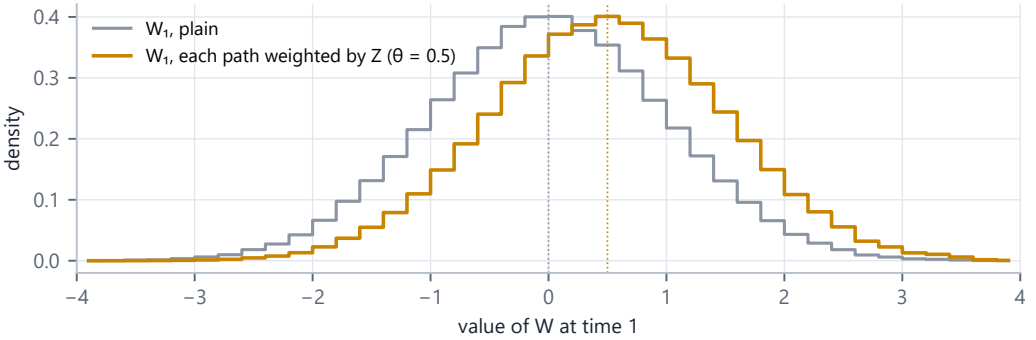


Figure 18.2. The distribution of W_1 (grey), and the same simulated values with each path weighted by $Z_1 = e^{(\theta W_1 - \theta^2/2)}$, $\theta = 0.5$ (amber). The weights move the whole distribution to the right by θ, without changing its shape.

Numbers, $\theta = 0.5$: $E[Z_1 W_1] = 0.5014$ (theory $\theta = 0.5$); and $E[Z_1 \mathbf{1}\{W_1 > 0.5\}] = 0.5004$, which is $P(W_1 + 0.5 > 0.5) = 0.50$, while without the weights $P(W_1 > 0.5) = 0.3085$.

What Girsanov's theorem adds (Lesson 25): not only W_T but the whole path is shifted. Under the reweighted probability, $\tilde{W}_t = W_t - \theta t$ is a Brownian motion. Choose $\theta = (\mu - r)/\sigma$, and the share's drift μ becomes r: the real world becomes the risk-neutral one, exactly as on the tree.

What to remember

- **Martingale test:** $dX = a\,dt + b\,dW$ is a martingale iff $a = 0$ (given $E\int b^2\,dt < \infty$). Constant expectation is not enough.
- **Exponential martingale:** $Z_t = e^{\theta W_t - \frac{1}{2}\theta^2 t}$, $dZ = \theta Z\,dW$, $E[Z_t] = 1$. The $-\frac{1}{2}\theta^2 t$ cancels Itô's correction.
- **Polynomial martingales:** $W^2 - t$, $W^3 - 3tW$, $W^4 - 6tW^2 + 3t^2$; they give the moments of W.
- **Product rule:** $d(XY) = X\,dY + Y\,dX + dX\,dY$; the last term vanishes if either has no dW part.
- $d(e^{-rt}S) = (\mu - r)e^{-rt}S\,dt + \sigma e^{-rt}S\,dW$: a martingale iff $\mu = r$.
- **Reweighting** by Z_T shifts W_T by θT: $E[Z_T f(W_T)] = E[f(W_T + \theta T)]$.

Common mistakes

- **"Constant mean, so a martingale."** W^3 has mean 0 for all t and is not a martingale.
- **Forgetting** $dX\,dY$ in the product rule when both processes depend on the same W.

- **Forgetting f_t** when checking $e^{\theta W - \frac{1}{2}\theta^2 t}$: without it the dt terms don't cancel.
- **"$e^{\theta W_t}$ has mean 1."** Its mean is $e^{\frac{1}{2}\theta^2 t}$.
- **Reweighting in the wrong direction.** Weighting by $e^{\theta W_T - \frac{1}{2}\theta^2 T}$ shifts W by $+\theta T$; to remove a drift of θ, weight by $e^{-\theta W_T - \frac{1}{2}\theta^2 T}$.

Exercises

Solutions are in *Part 3: Solutions*.

18.1 For each process, find the dt term and decide whether it is a martingale: W_t^3, $W_t^3 - 3tW_t$, e^{W_t}, $e^{W_t - t/2}$, tW_t.

18.2 Find the constant c that makes $e^{2W_t - ct}$ a martingale. More generally, for which c is $e^{\theta W_t - ct}$ a martingale?

18.3 Use the product rule to find $d(tW_t)$, and show that $tW_t - \int_0^t W_s\,ds$ is a martingale. Deduce $E\big[\int_0^1 W_s\,ds\big] = 0$.

18.4 Show that $W_t^4 - 6tW_t^2 + 3t^2$ is a martingale, and use it to find $E[W_t^4]$.

18.5 Let W and B be independent Brownian motions and $X = e^{W_t - t/2}$, $Y = e^{B_t - t/2}$. Use the product rule (with $dW\,dB = 0$) to show XY is a martingale. What changes if W and B have correlation ρ?

18.6 For $dS = \mu S\,dt + \sigma S\,dW$, use the product rule to find $d(e^{-rt}S_t)$, and show it is a martingale only if $\mu = r$. Compute $E[e^{-r}S_1]$ for $\mu = 8\%$.

18.7 *(Code)* Test the martingale property of W^3 and $W^3 - 3tW$ by grouping simulated paths by $W_{0.5}$, as in the lesson. Which one shows a gap, and how does the gap depend on the group?

18.8 *(Code)* Check $E[Z_T \mathbf{1}\{W_T > a\}] = P(W_T + \theta T > a)$ for $\theta = 0.5$, $T = 1$ and $a = 0, 0.5, 1$.

Self-test

1. What c makes $e^{0.3W_t - ct}$ a martingale?
2. W_t^3 has mean 0 at every time. Why is it not a martingale?
3. What is $d(XY)$ when X is a smooth function of time?

Companion notebook: `s5e18_ito_toolbox.ipynb`. Every number in this chapter is computed by it.

Formula sheet and glossary

Formula sheet

Brownian motion (Lesson 13)

- $W_0 = 0$; independent increments; $W_t - W_s \sim N(0, t-s)$; continuous paths.
- $E[W_t] = 0$, $\mathrm{Var}(W_t) = t$, $\mathrm{Cov}(W_s, W_t) = \min(s,t)$.
- Scaled random walk: n steps of $\pm 1/\sqrt{n}$ per unit time $\to W$.
- Self-similarity: $W_{ct}/\sqrt{c}$ is a Brownian motion.
- Martingales: W_t, $W_t^2 - t$, $e^{\sigma W_t - \frac{1}{2}\sigma^2 t}$.

Paths (Lesson 14)

- Slope over h: $N(0, 1/h)$; no derivative.
- $E\sum|\Delta W_i| = \sqrt{2nT/\pi} \to \infty$; $\sum(\Delta W_i)^2 \to T$, variance $2T^2/n$.
- $dt\,dt = 0$, $dW\,dt = 0$, $dW\,dW = dt$, $dW\,dB = \rho\,dt$.
- $[\ln S]_T = \sigma^2 T$ (realised variance).

The Itô integral (Lesson 15)

- $\int_0^T H\,dW = \lim \sum H_{t_i}\Delta W_i$ (left endpoints, adapted H).
- $E\int H\,dW = 0$; $E[(\int H\,dW)^2] = E\int H^2\,dt$; $[\int H\,dW]_T = \int H^2\,dt$.
- $\int_0^T W\,dW = \frac{1}{2}W_T^2 - \frac{1}{2}T$.
- $\int_0^T f(t)\,dW_t \sim N(0, \int f^2\,dt)$; $E[\int f\,dW \int g\,dW] = \int fg\,dt$.

Itô's lemma (Lesson 16)

- $dX = a\,dt + b\,dW \Rightarrow (dX)^2 = b^2\,dt$.
- $df(W) = f'(W)\,dW + \frac{1}{2}f''(W)\,dt$.
- $df(t,X) = (f_t + af_x + \frac{1}{2}b^2 f_{xx})\,dt + bf_x\,dW$.
- $\frac{d}{dt}E[f(W_t)] = \frac{1}{2}E[f''(W_t)]$; $E[e^{W_t}] = e^{t/2}$, $E[W_t^4] = 3t^2$, $E[W_t^6] = 15t^3$, $E[\cos W_t] = e^{-t/2}$.

Geometric Brownian motion (Lesson 17)

- $dS = \mu S\,dt + \sigma S\,dW$; $d\ln S = (\mu - \frac{1}{2}\sigma^2)\,dt + \sigma\,dW$.
- $S_t = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}$; mean $S_0 e^{\mu t}$, median $S_0 e^{(\mu - \frac{1}{2}\sigma^2)t}$.

- $E[S_t^2] = S_0^2 e^{(2\mu+\sigma^2)t}$.
- Euler: $S_{k+1} = S_k(1 + \mu\Delta t + \sigma\Delta W_k)$; strong order ½, weak order 1. Milstein adds $\frac{1}{2}\sigma^2(\Delta W_k^2 - \Delta t)$: strong order 1.

The toolbox (Lesson 18)

- $dX = a\,dt + b\,dW$ is a martingale iff $a = 0$ (given $E\int b^2\,dt < \infty$).
- $Z_t = e^{\theta W_t - \frac{1}{2}\theta^2 t}$: $dZ = \theta Z\,dW$.
- $d(XY) = X\,dY + Y\,dX + dX\,dY$.
- $d(e^{-rt}S) = (\mu - r)e^{-rt}S\,dt + \sigma e^{-rt}S\,dW$.
- $E[Z_T f(W_T)] = E[f(W_T + \theta T)]$.

The course's numbers

Share price today, S_0	100
Strike, K	105
Risk-free rate, r (continuous)	3%
Volatility, σ	20%
Expected return in Part 3, μ	3% (equal to r)
Horizon, T	1 year

Glossary

Adapted. Known at each time from the path up to that time.

Brownian motion (Wiener process). A continuous random path starting at 0 with independent, normal increments of variance equal to the elapsed time.

Diffusion coefficient (volatility). The b in $dX = a\,dt + b\,dW$.

Drift. The a in $dX = a\,dt + b\,dW$: the predictable part of the move.

Euler (Euler-Maruyama) method. Simulating $dX = a\,dt + b\,dW$ by $X_{k+1} = X_k + a\Delta t + b\Delta W_k$.

Exponential martingale. $e^{\theta W_t - \frac{1}{2}\theta^2 t}$; more generally $e^{\int \theta\,dW - \frac{1}{2}\int \theta^2\,dt}$.

First variation. The limit of $\sum |\Delta X_i|$: the length of a path. Infinite for Brownian motion.

Geometric Brownian motion. $dS = \mu S\,dt + \sigma S\,dW$; the Black-Scholes model of a share.

Itô integral. $\int H\,dW$, the limit of left-endpoint sums for an adapted H; a martingale.

Itô isometry. $E[(\int H\,dW)^2] = E\int H^2\,dt$.

Itô process. $dX = a\,dt + b\,dW$, that is, $X_t = X_0 + \int a\,ds + \int b\,dW$.

Itô's lemma. The chain rule for Itô processes, with the extra term $\frac{1}{2}f_{xx}(dX)^2 = \frac{1}{2}b^2 f_{xx}\,dt$.

Milstein scheme. Euler plus the term $\frac{1}{2}bb'(\Delta W^2 - \Delta t)$; strong order 1.

Multiplication table. $dt\,dt = dW\,dt = 0$, $dW\,dW = dt$.

Product rule (Itô). $d(XY) = X\,dY + Y\,dX + dX\,dY$.

Quadratic variation. The limit of $\sum(\Delta X_i)^2$; $[W]_T = T$, $[\int H\,dW]_T = \int H^2\,dt$.

Realised variance. $\sum(\Delta \ln S)^2$ over observed prices; tends to $\sigma^2 T$.

Self-similarity. $W_{ct}/\sqrt{c}$ has the same law as W_t: Brownian paths look alike at every scale.

Stratonovich integral. The limit of sums using the average of the two endpoints; follows ordinary calculus rules but is not adapted.

Strong and weak error. The average distance between a simulated and an exact path at T (strong), and the distance between their averages (weak).

Further reading

- Steven E. Shreve, *Stochastic Calculus for Finance II: Continuous-Time Models*, Chapters 3 and 4: Brownian motion and Itô calculus with full rigour.
- Thomas Mikosch, *Elementary Stochastic Calculus with Finance in View*, Chapters 1 to 3: a gentle, careful companion to Part 3.
- Peter E. Kloeden and Eckhard Platen, *Numerical Solution of Stochastic Differential Equations*: everything about Euler, Milstein and beyond.